



GE Medical Systems

Technical Publications

Direction 5133318

Revision 6

Signa® 3.0T with EXCITE Technology System Upgrade from 11.x to 12.x

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
http://data.supportcentral.ge.com/upload/15325/doc_324531.doc
- Fill out a report on <http://3.28.216.127/sctq/InstallFulfill/IFHome.htm>
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003



GE Medical Systems

GE Medical Systems: Telex 3797371
P.O. Box 414, Milwaukee, Wisconsin 53201 U.S.A.
(Asia, Pacific, Latin America, North America)

GE Medical Systems — Europe: Telex 261794
Shortlands, Hammersmith, London W6 8BX U.K.

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1	May 4, 2005	Initial Release.
2	Dec 13, 2005	Sec 4 – Corrected GOC Power Strip connection designators on cable maps. Sec 6 – Updated instructions by adding note about loosening Longitudinal Motor Drive asm for easier removal of old bridge and installation of new bridge (MRIhc11778).
3	Aug 11, 2006	Sec 6 – Removed note in 6–4–1–4, updated the illustration in Section 6–4–1–5 to show a rear bridge with two cable track troughs (MRIhc11777) and changed the fastener in Section 6–4–2–1 to a 10–32 from a M6 for fastening the 16 Channel switch mounting plate. Sec 9 – Revised the instructions for installing new boards in RRF Chassis and added the alternative RF–DIF board, 5250028.
4	Oct 26, 2006	Section 6 – Updated PAC replacement instructions with information on removing the PAC Control Housing (MRIhc17607, MRIhc18836).
5	Mar 7, 2007	Section 6: Removed PAC 2B part number. Section 9: Updated instructions for installing 8 Channel upgrade. Revised 16 Channel installation procedures by updating MUX board part number and note for recommendation on existing MUX board. Added instructions for storing Service Cable Kit.
6	Jun 17, 2009	Section 6: Removed Section 6–6, Install Rating Plate on Front Cover (PQR 13246367).

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* This revision number/letter corresponds to the indicated document's revision control system.

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SECTION 1 – GETTING STARTED

1-1 UPGRADE INTRODUCTION

This publication documents the mechanical installation information for upgrading a Signa® 3.0T with EXCITE Technology (3T/94) to Signa 3.0T EXCITE HD (Release 12.x). Refer to illustration 1-1 for pictorial overview of upgraded system.



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

Tools and test equipment to install and calibrate the Signa 3.0T EXCITE HD system are listed in Direction 5133305, *Signa 3.0T EXCITE Upgrade Pre-Installation* manual, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46-301450G1 Fiber Optic Cable Connector Repair kit

1-3 PRE-DELIVERY FDO CHECKS FOR SIGNA EXCITE

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa 3.0T Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1-4 UPGRADE SCHEDULE CONSIDERATIONS

FMI's, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1-5 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short. Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1-6 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1-800-548-3366 and use option 8.
http://data.supportcentral.ge.com/upload/15325/doc_324531.doc
- Fill out a report on <http://3.28.216.127/sctq/InstallFulfill/IFHome.htm>
- Contact your local service coordinator for more information on this process.

1-7 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-7-1 MATERIAL POLICY OVERVIEW

1-7-1-1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1-7-1-2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-7-2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment deinstalled for the EXCITE HD (Release 12.x) upgrade to GoldSeal which includes:

- Driver Module
- SRI 2
- 8 Channel Switch
- LPCA
- Bridge
- Cable Take-Up
- Exciter / Receiver Boards

1-7-3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.
- Packing material

1-7-4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for deinstallation procedures of site.

1-7-5 REGION SPECIFIC RETURN INFORMATION**1-7-5-1 Sites in Americas**

A GoldSeal return pre-printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1-877-251-4468 or 262-513-4147

1-7-5-1 Sites in Americas (Continued)

For non-GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)-747-6997 or Dial Comm 8*579-6997

Fax: (414)-747-6855

Field ASPEN: 1-800-525-1516, Box #26041

1-7-5-2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS-Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1-7-5-3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS-E – GOLD SEAL OPERATION & RECYCLING CENTER

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL

Quai 45 à 49 – 91090 LISSES – FRANC

Tel: (+33) 1-60-91-57-35 or (+33) 1 64 97 55 41

Fax: (+33) 1-60-86-12-31

For non-Gold Seal returns in Europe:

European Renewable Resources Center

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2

Quai 45 à 49 – 91090 LISSES – FRANCE

TEL = (+33)-1-64-97-55-41 FAX = (+33)-1-60-86-12-31

1-7-6 PRODUCT LOCATOR INFORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1-8.

1-8 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at:
<http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following:
[Product Locator Support Central Page](http://supportcentral.ge.com/15563) – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-9 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

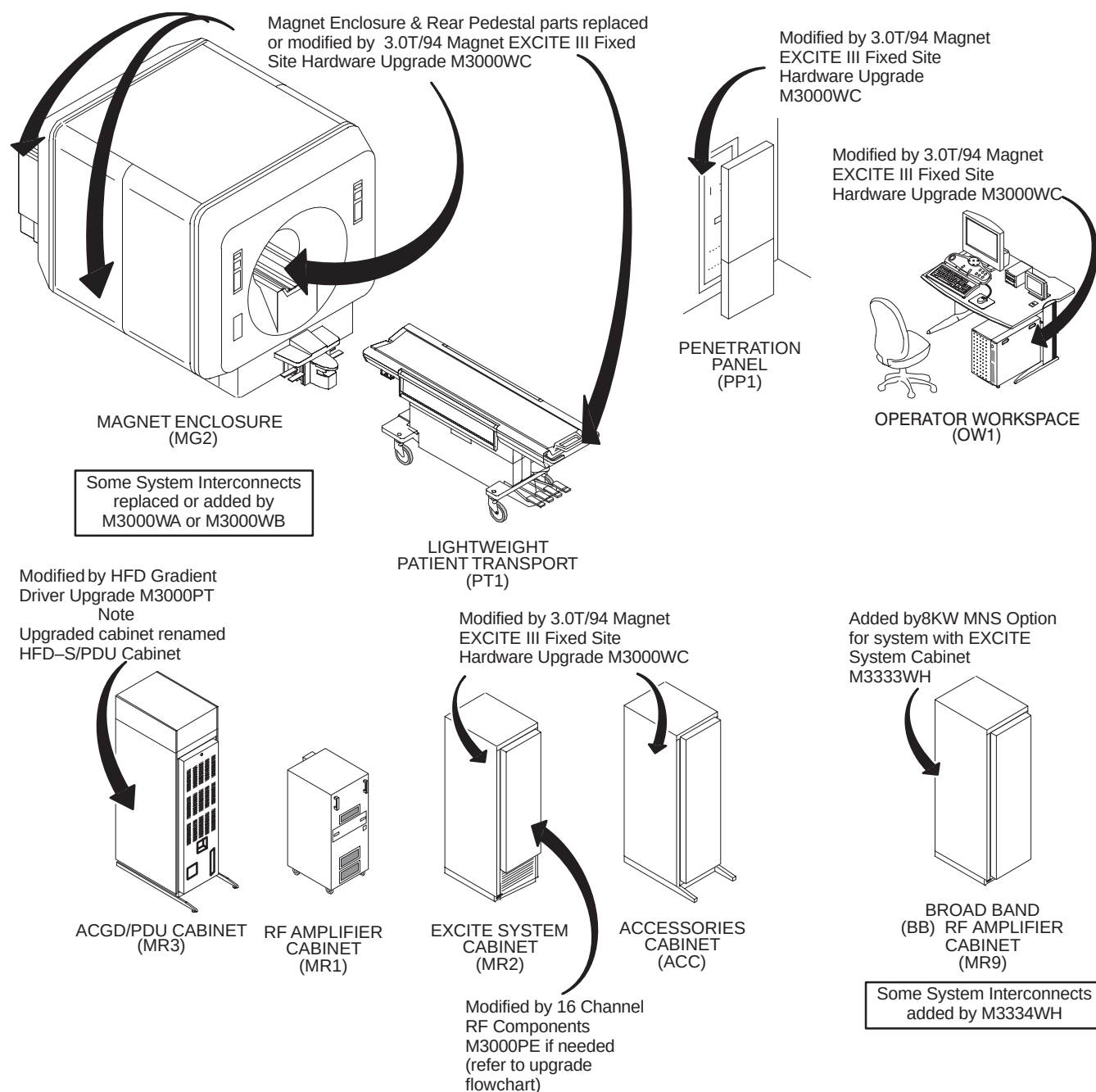
- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

Aside: Removing and installing cables is more efficient and accurate when the equipment room is un-cluttered. It is strongly recommended that this be done after old equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1-10 UPGRADE SYSTEM CONFIGURATION

Illustration 1-1 shows the major equipment replaced and modified by Signa 3.0T (3T/94) EXCITE 11.x to 12.x Upgrade.



SIGNA 3T/94 E2 EXISTING HARDWARE REPLACED OR MODIFIED UPGRADE TO EXCITE HD (Release 12.x)

ILLUSTRATION 1-1

SECTION 2 – UPGRADE INSTALLATION PROCESS

2-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

2-2 UPGRADE PROCEDURE

The following upgrade flow charts contain information for upgrading to Signa EXCITE HD 3.0T (Release 12.x).

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

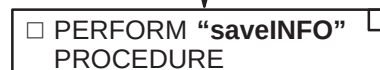
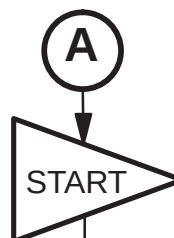
This upgrade flowchart has been developed assuming that all Signa EXCITE HD equipment has been delivered together. Ensure that every part required is available before starting.

2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART

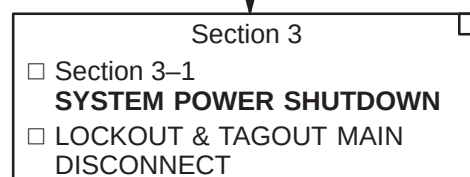
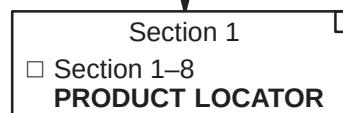
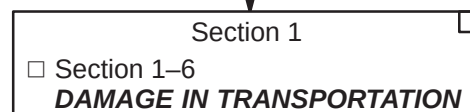
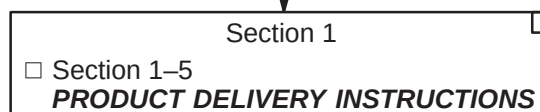
Procedures to be Completed by Field Engineer



IMPORTANT : PER ISO
9001 REQUIREMENTS

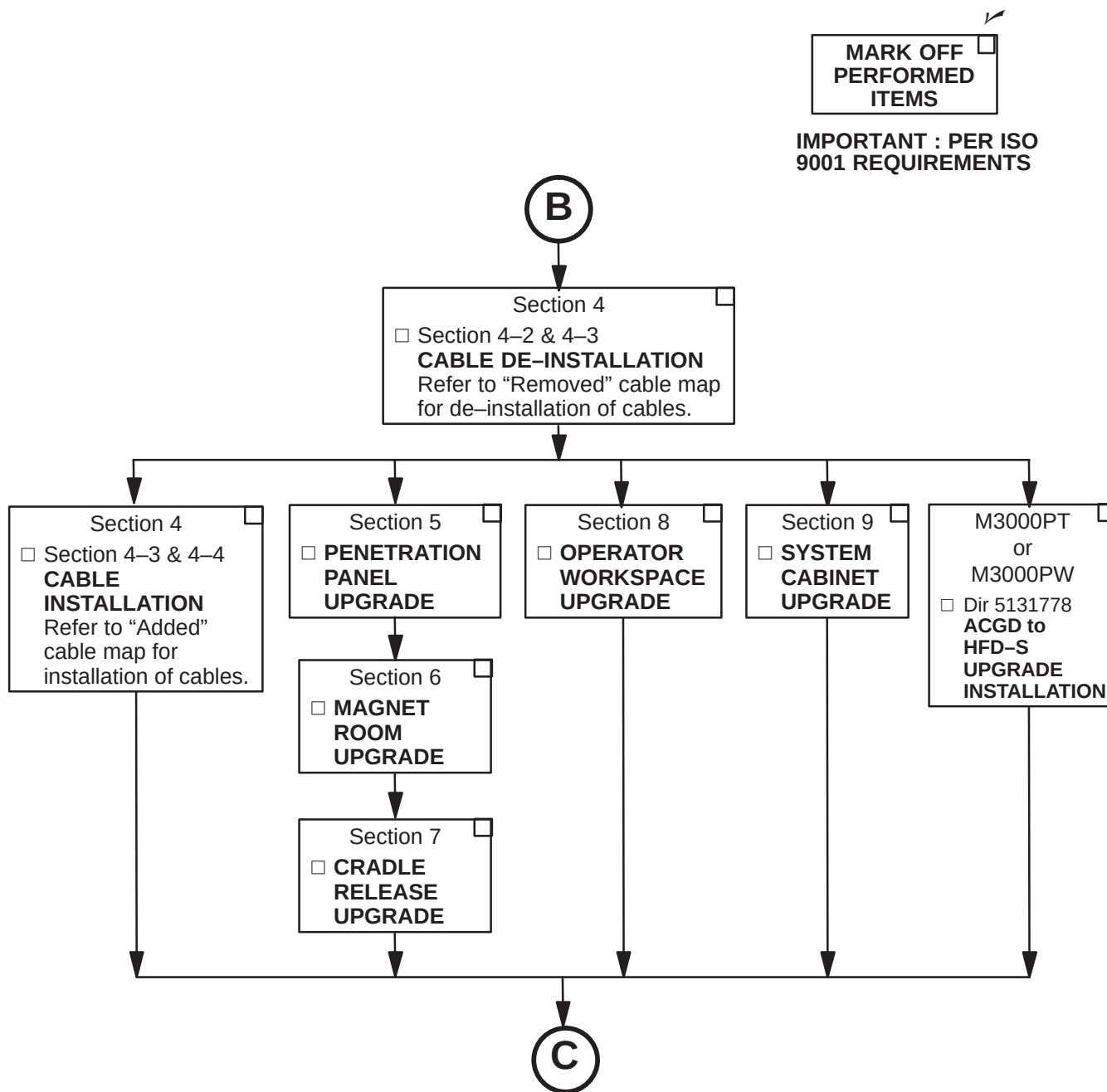


Procedures to be Completed by FE and/or Mechanical Installers



2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers

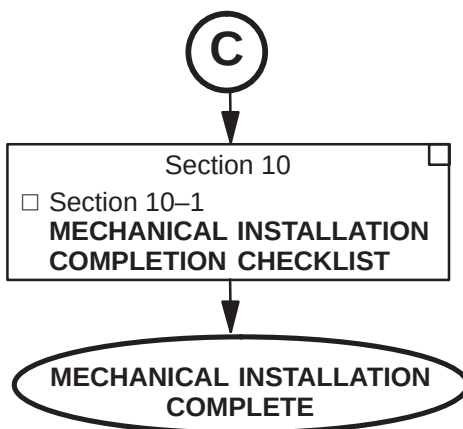


2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

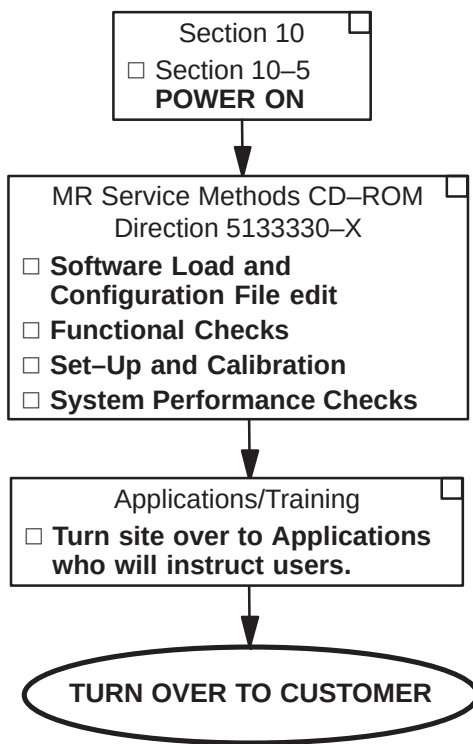
Procedures to be Completed by FE and/or Mechanical Installers


**MARK OFF
PERFORMED
ITEMS**

**IMPORTANT : PER ISO
9001 REQUIREMENTS**



Procedures to be Completed by FE and Applications



SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Make sure that site customer has archived all patient information. Execute “Save Info”.
2. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the FULL OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 4 – SYSTEM UPGRADE CABLES

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4-1 INTRODUCTION

This section provides information on removing, relabeling and installing system cables for upgrading to Signa 3.0T (3T/94) EXCITE HD, Release 12.x.

4-2 CABLES DE-INSTALLED FOR UPGRADING TO EXCITE HD

Table 4-1 lists all cables to be removed from 3T/94 with EXCITE (11.x) system configurations upgrading to a Signa EXCITE HD (12.x) system. Use table along with "Affected Cables" interconnect map for removal information.

TABLE 4-1
3T/94 with EXCITE CABLES REMOVED FOR UPGRADE TO 3T EXCITE HD

RUN#	FROM	EQUIPMENT	TO	EQUIPMENT
231	MR2-A11-J2	System Cabinet	PP1-J8	Pen Panel
258	MG3-A17-J1	Rear Pedestal	PP1-J4	Pen Panel
262	MG3-A3-A5-OUTPUT	BodyCoil Preamp	PP1-J8	Pen Panel
404	MG3-A17-J1	Rear Pedestal	MG2-A1- A5-J6	LPCA
488	PP1-J80	Pen Panel	MR2-A11-J8	System Cabinet
489	PP1-J81	Pen Panel	MR2-A11-J9	System Cabinet
490	PP1-J82	Pen Panel	MR2-A11-J10	System Cabinet
491	PP1-J80	Pen Panel	MG3-A36-J7	8 Channel Sw
492	PP1-J81	Pen Panel	MG3-A36-J8	8 Channel Sw
493	PP1-J82	Pen Panel	MG3-A36-J9	8 Channel Sw
499	PP1-A14-J84	Pen Panel	MR2-A11-J5	System Cabinet
728	MG3-A36-J11	8 Channel Sw	MG2-A16-OUT-J1	LPCA
729	MG3-A36-J12	8 Channel Sw	MG2-A16-OUT-J2	LPCA
730	MG3-A36-J13	8 Channel Sw	MG2-A16-OUT-J3	LPCA
731	MG3-A36-J14	8 Channel Sw	MG2-A16-OUT-J4	LPCA
732	MG3-A36-J15	8 Channel Sw	MG2-A16-OUT-J5	LPCA
733	MG3-A36-J16	8 Channel Sw	MG2-A16-OUT-J6	LPCA
734	MG3-A36-J17	8 Channel Sw	MG2-A16-OUT-J7	LPCA
735	MG3-A36-J18	8 Channel Sw	MG2-A16-OUT-J8	LPCA
749	MG2-A33-J10	SRI Module	MG2-A36	Temp Sensor
1011	MG2-A33-J5	SRI Module	MG2-A4-J4	Rt Oper Cntrl
1035	MG3-A36-J21	8 Channel Sw	MG2-A16-A3/A5-J4	LPCA
1042	PP1-A16-J77	Pen Panel	MG3-A36-J5	8 Channel Sw
1049	MG3-A36-J19	8 Channel Sw	MG2-A16-A8-J9	LPCA
1053	MR2-A11-J54	System Cabinet	PP1-J85	Pen Panel
1054	MG3-A36-J20	8 Channel Sw	MG2-A33-J15	SRI Module
1057	PP1-J85	PenPanel	MG3-A8-J1	UTNS Antenna
1078	MG3-A36-J29,J30	8 Channel Sw	MG2-A33-J18,J19	SRI Module

4-2 CABLES DE-INSTALLED FOR UPGRADING TO SIGNA EXCITE HD (Continued)TABLE 4-1 (Continued)
3T/94 with EXCITE CABLES REMOVED FOR UPGRADE TO 3T EXCITE HD

RUN#	FROM	EQUIPMENT	TO	EQUIPMENT
1153	MG3-A17-J2	Rear Pedestal	MG2-A16-A3 OUT	LPCA
1155	MG3-A36-J28	8 Channel Sw	MG2-A16-J2	LPCA
1168	MR2-A11-J40	System Cabinet	PP1-J6	PenPanel
1169	MR2-A11-J41	System Cabinet	PP1-J7	PenPanel
1170	MR2-A11-J42	System Cabinet	PP1-J58	PenPanel
1171	MR2-A11-J43	System Cabinet	PP1-J59	PenPanel
1172	MG3-A36-J1	8 Channel Sw	PP1-J6	PenPanel
1173	MG3-A36-J2	8 Channel Sw	PP1-J7	PenPanel
1174	MG3-A36-J3	8 Channel Sw	PP1-J58	PenPanel
1175	MG3-A36-J4	8 Channel Sw	PP1-J59	PenPanel
1176	MG3-A36-J6	8 Channel Sw	PP1-J84	PenPanel

4-3 CABLE MAPS

A double set of "Removed" and "Added" cables maps is provided. One set for the equipment room and the other for the magnet room. Tape maps in a convenient location to check off affected "Removed" and "Added" cables.

The Affected "Removed" Cable maps are located on the following pages.

- Page 4-5 Affected (removed) Cables for Upgrading to Signa 3.0T EXCITE HD (Equip. Room)
- Page 4-7 Affected (removed) Cables for Upgrading to Signa 3.0T EXCITE HD (Magnet Room)

The Affected "Added" Cable maps are located on the following pages.

- Page 4-9 Affected (added) Cables for Upgrading to Signa 3.0T EXCITE HD (Equip. Room)
- Page 4-11 Affected (added) Cables for Upgrading to Signa 3.0T EXCITE HD (Magnet Room)

Upon completion of connecting cables, return fold-out map to the binder for future reference.

4-4 SORTING AND ROUTING NEW CABLES

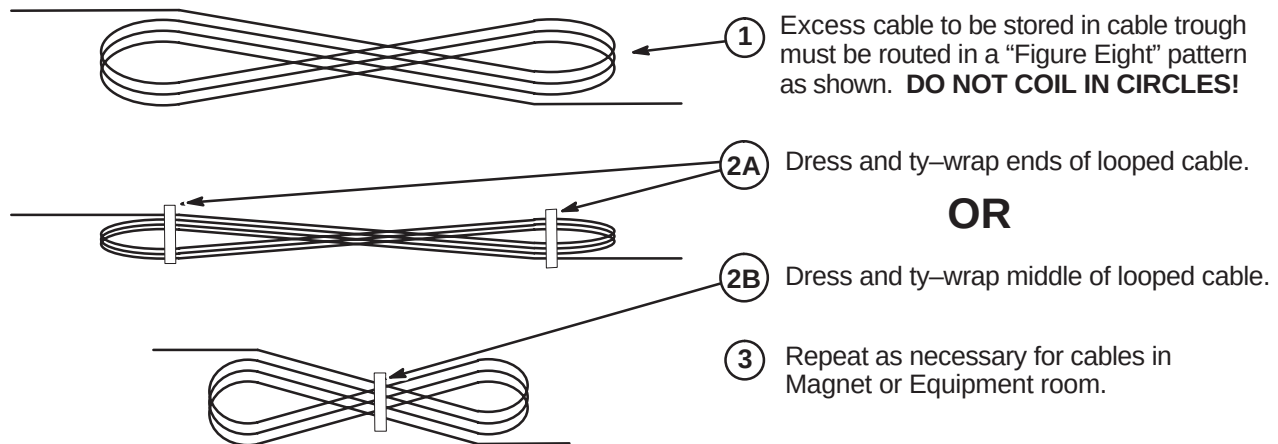
Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

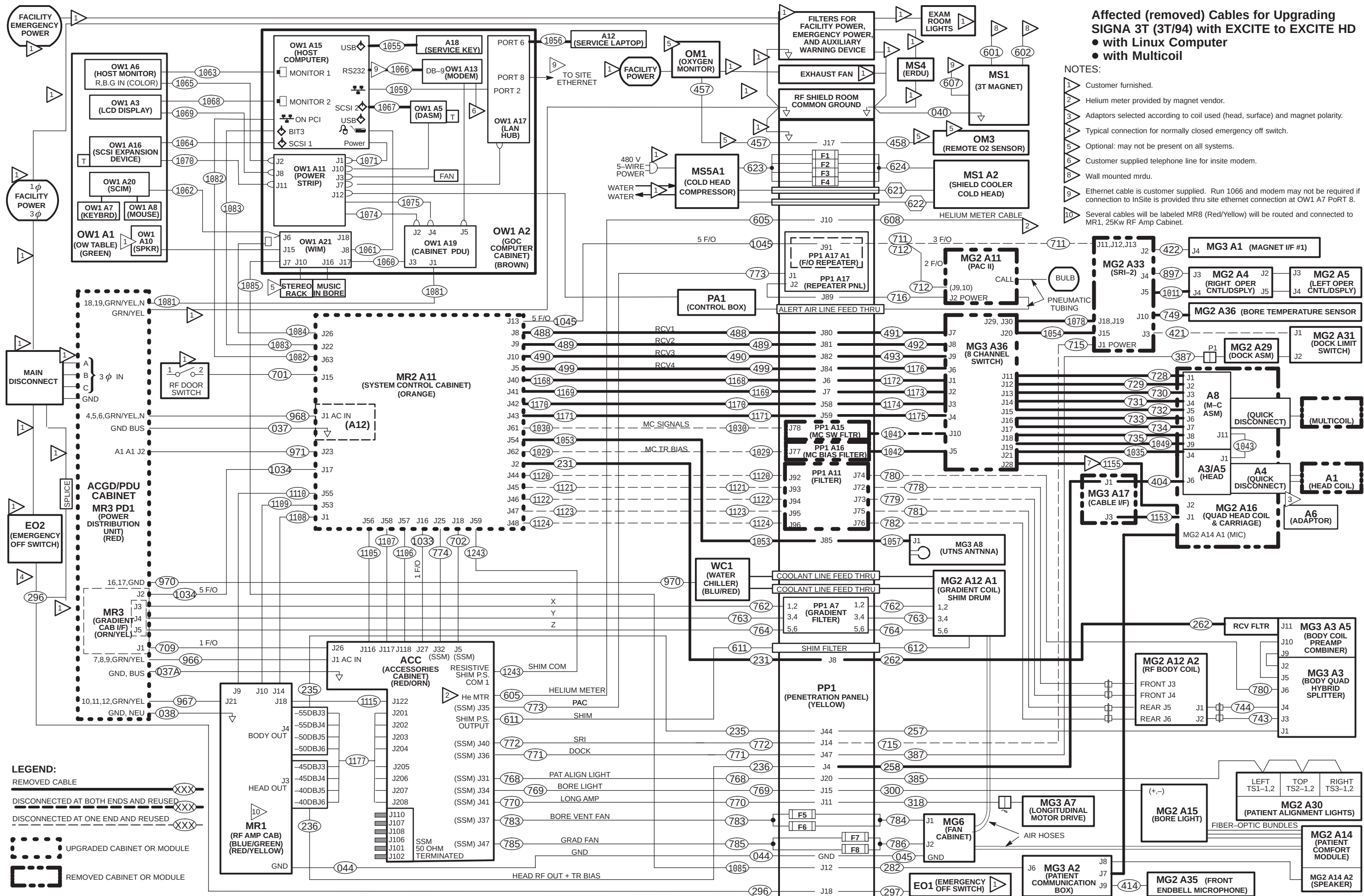
2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 4-1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber-optic cables.



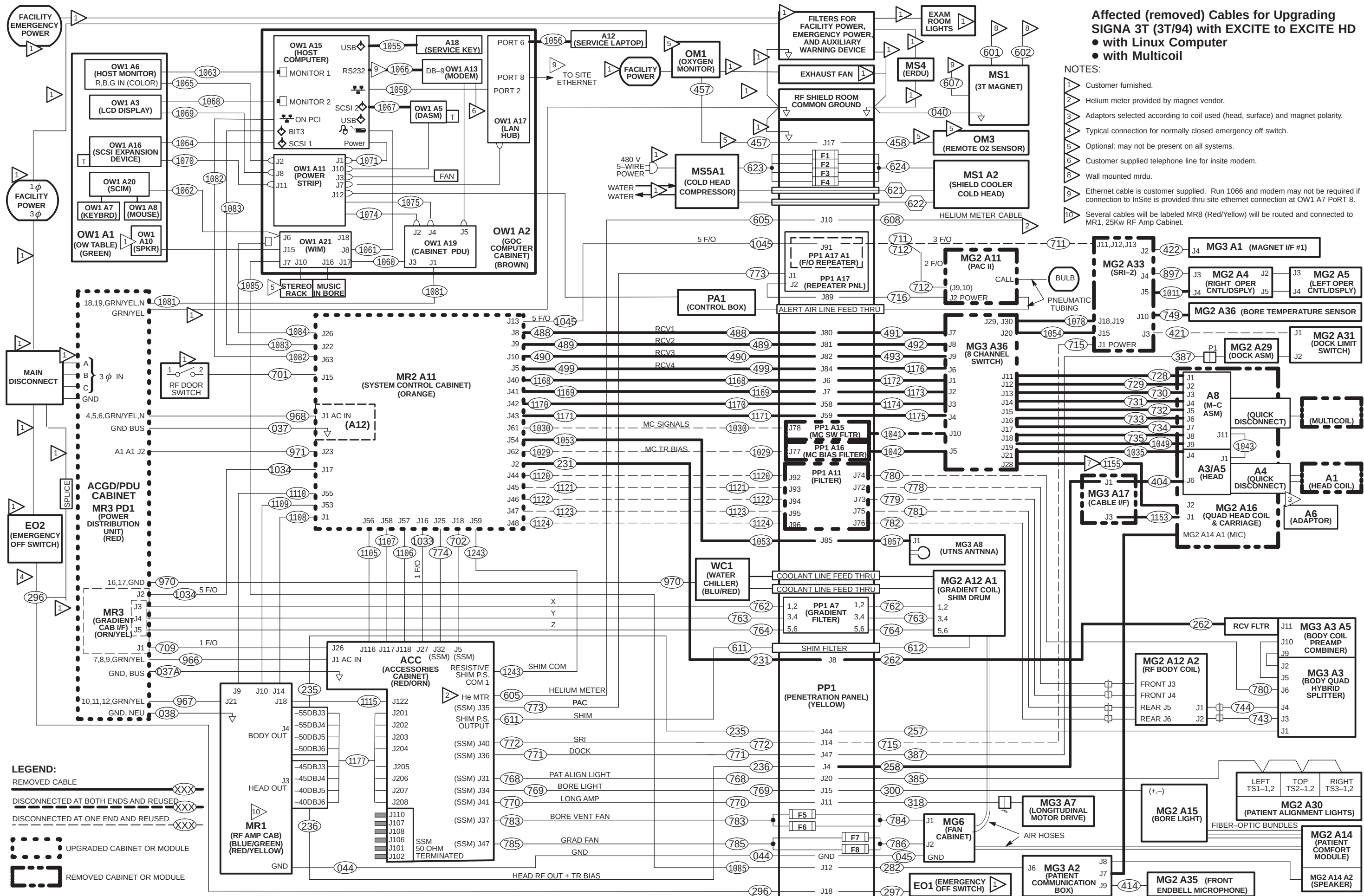
PROPER STORAGE OF EXCESS CABLES

ILLUSTRATION 4-1

4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.



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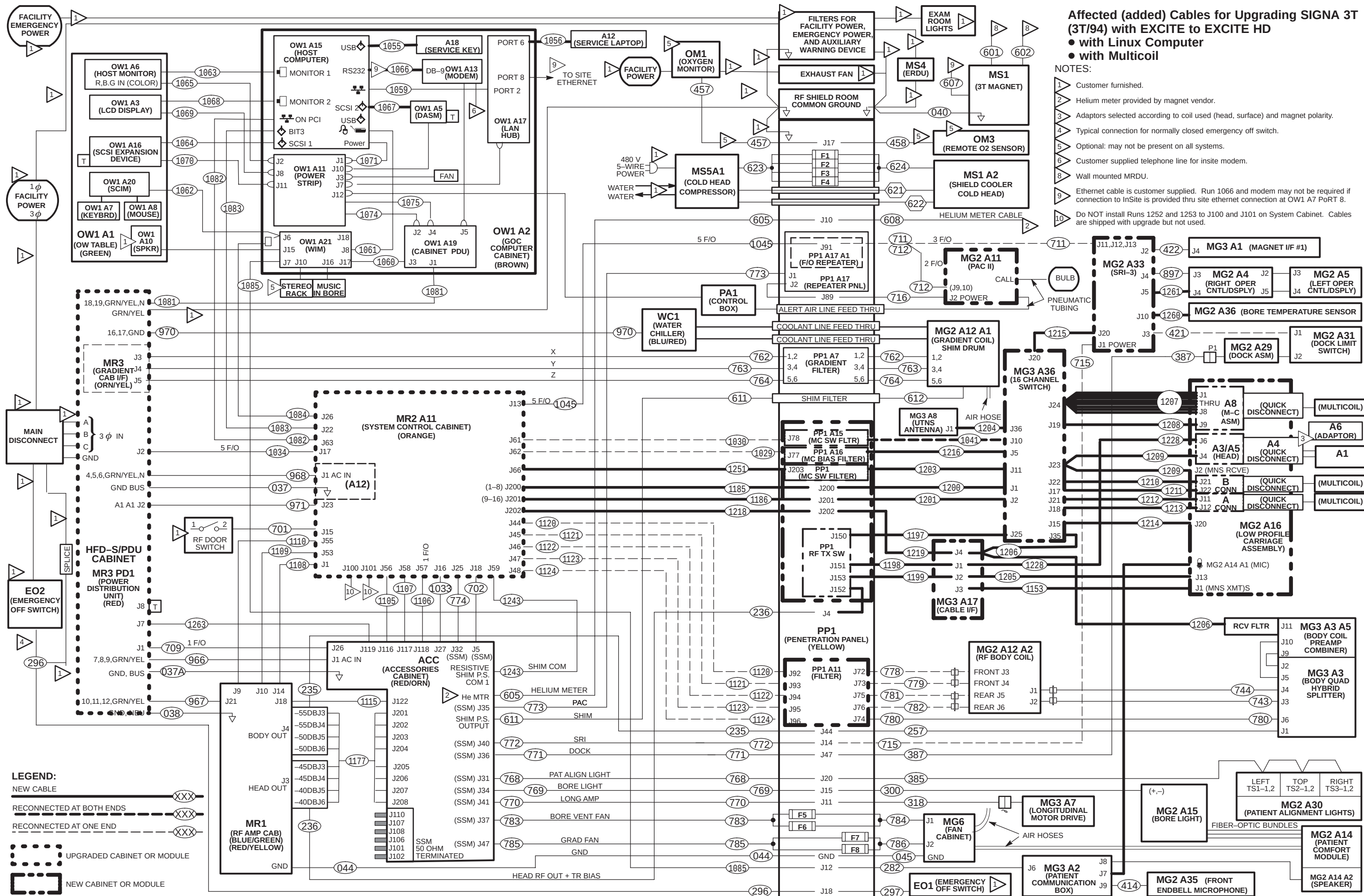


**Affected (removed) Cables for Upgrading
SIGNA 3T (3T/94) with EXCITE to EXCITE HD**

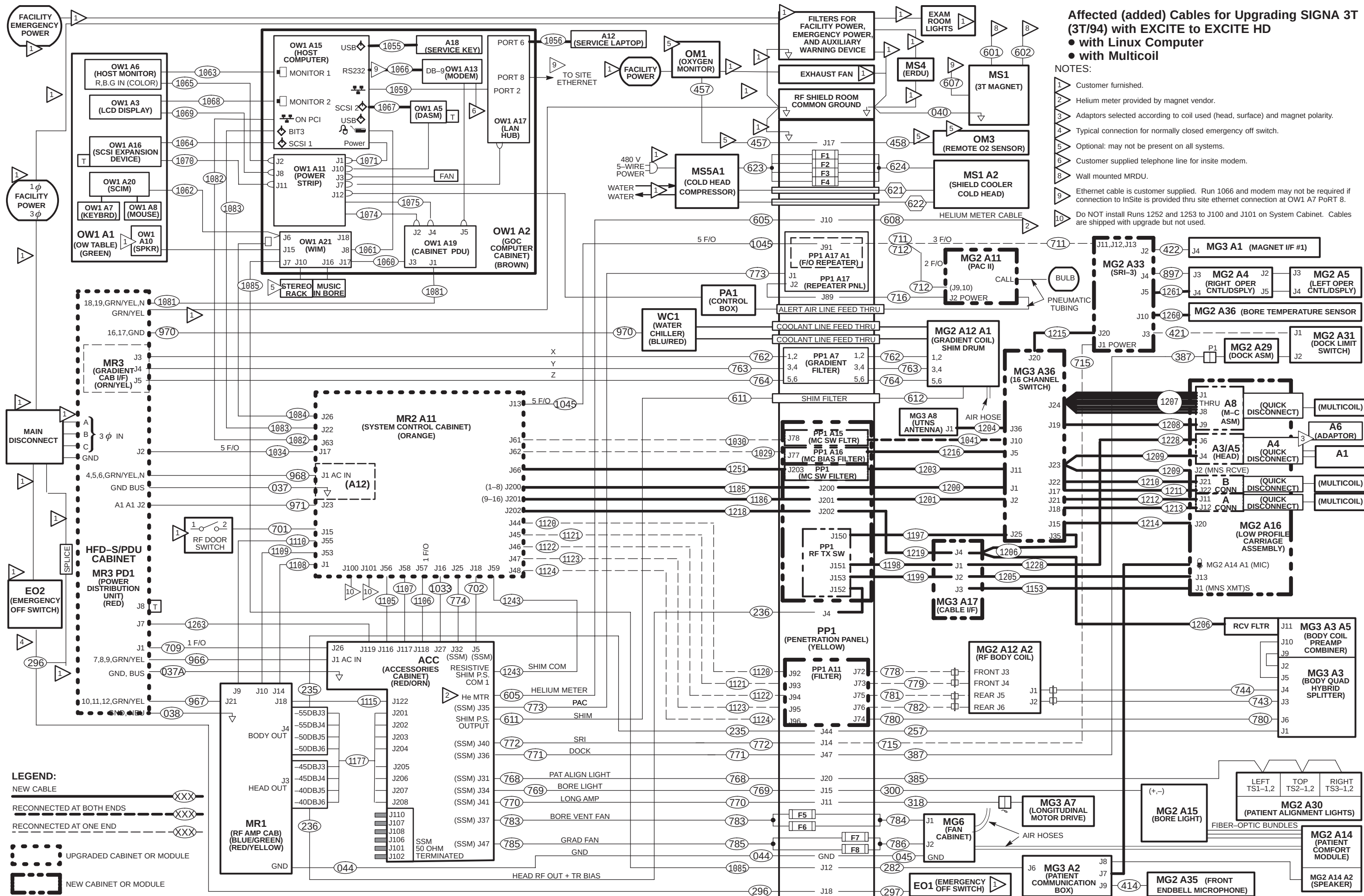
- with Linux Computer
- with Multicoil

- NOTES:
- 1 Customer furnished.
 - 2 Helium meter provided by magnet vendor.
 - 3 Adaptors selected according to coil used (head, surface) and magnet polarity.
 - 4 Typical connection for normally closed emergency off switch.
 - 5 Optional: may not be present on all systems.
 - 6 Customer supplied telephone line for insite modem.
 - 8 Wall mounted mrd.
 - 9 Ethernet cable is customer supplied. Run 1066 and modem may not be required if connection to InSite is provided thru site ethernet connection at OW1 A7 PoRt 8.
 - 10 Several cables will be labeled MR8 (Red/Yellow) will be routed and connected to MR1, 25Kw RF Amp Cabinet.

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SECTION 5 – PENETRATION PANEL

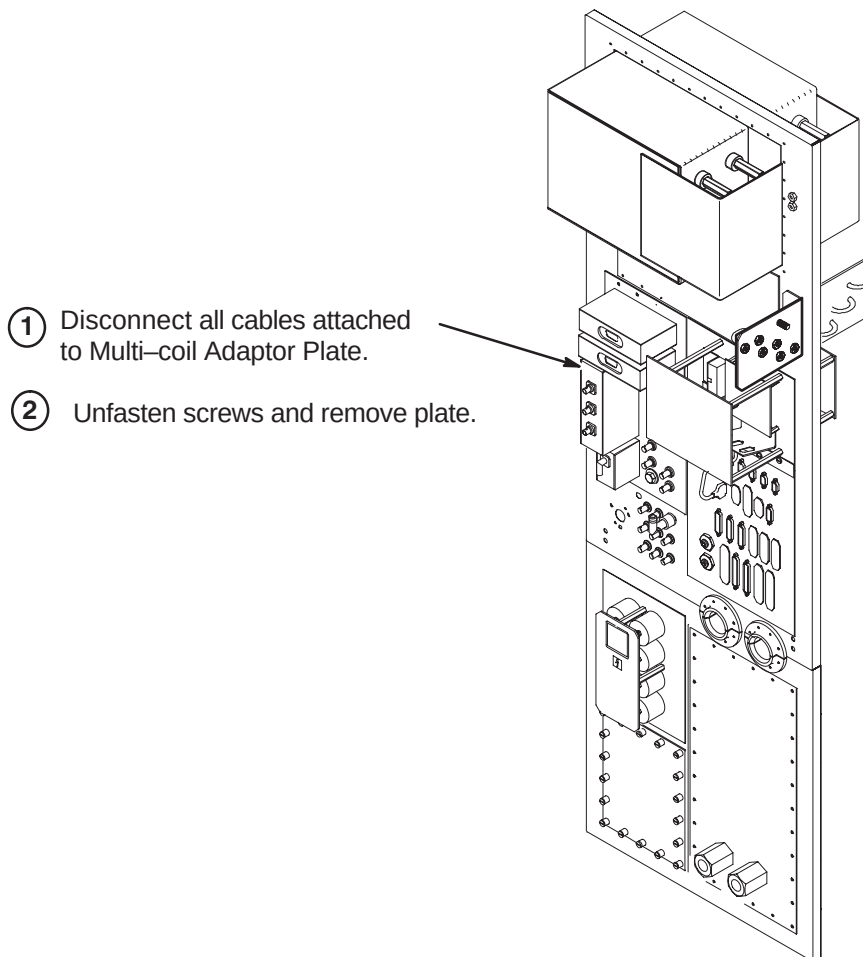
CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

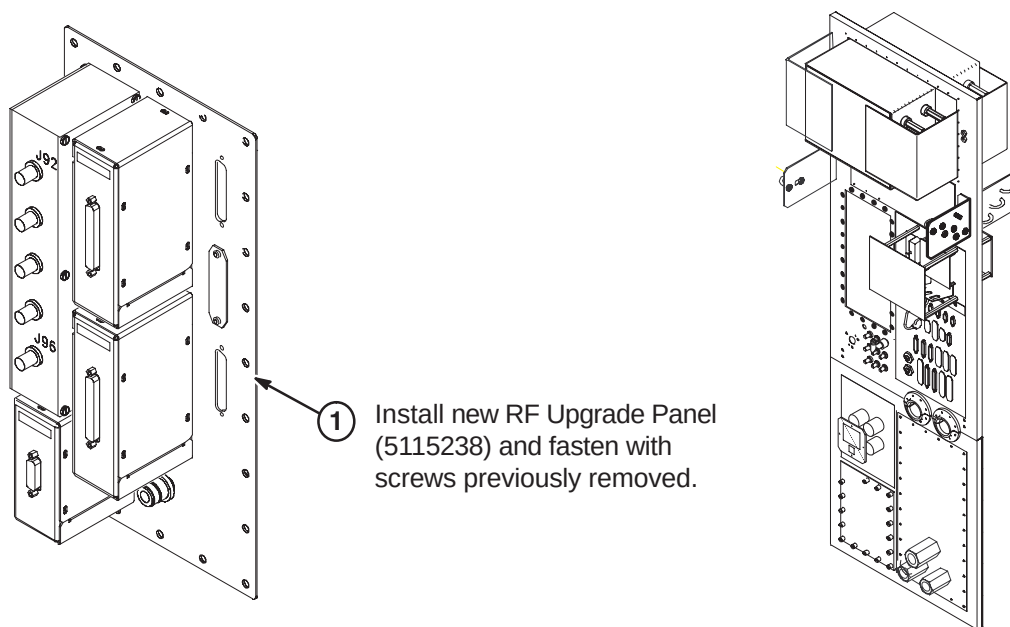
5-1 INTRODUCTION

The Multi-coil Adaptor Plate will be replaced by the RF Upgrade Panel (5115238).

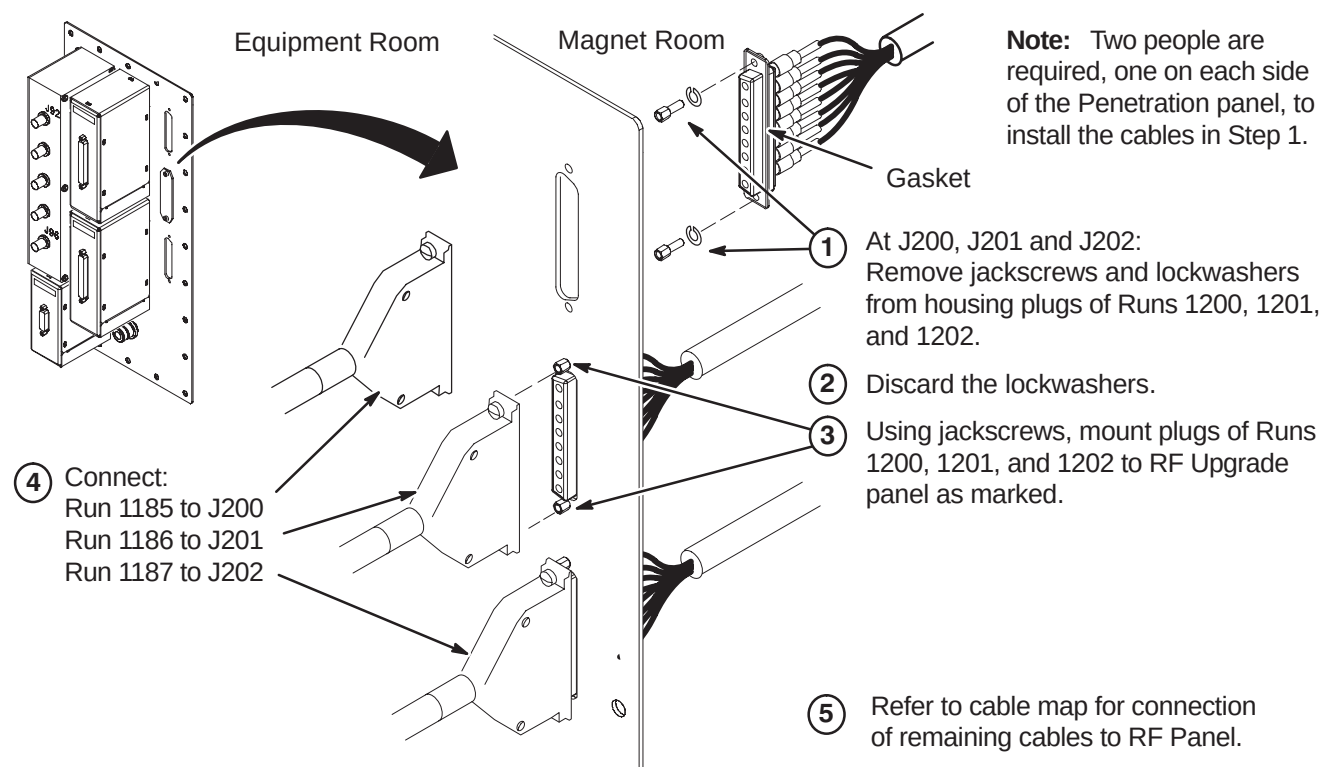
5-2 REMOVE MULTI-COIL ADAPTOR PLATE FROM PENETRATION PANEL



5-3 INSTALL NEW EXCITE RF PANEL



5-4 INSTALLATION OF CABLES AT J200 AND J201



SECTION 6 – MAGNET ROOM

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6-4-3 INSTALL LPCA AND CABLE TRACK	6-16
6-4-4 CONNECT REMAINING CABLES TO REAR PEDESTAL	6-21
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6-5-2 ROUTE, TERMINATE, AND CONNECT RF CABLES	6-25

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CAUTION

MAKE SURE POWER TO SYSTEM IS OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

6-1 INTRODUCTION

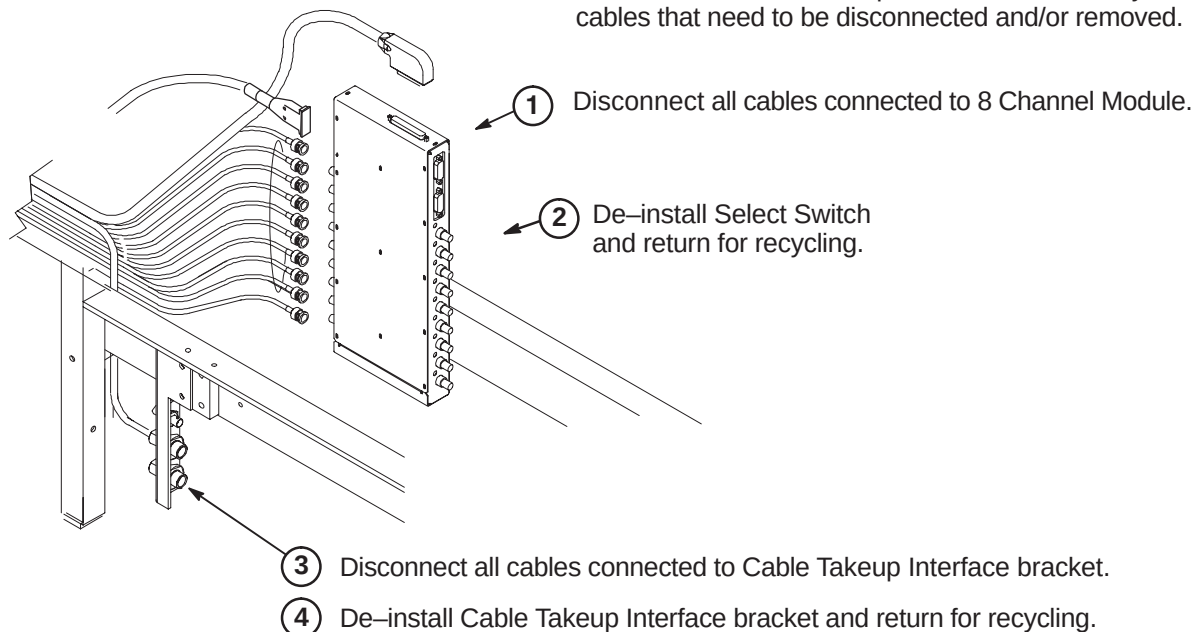
The Bridge, LPCA and Cable Track, SRI, PAC, and modules on the Rear Pedestal will be replaced.

Several cables will be removed and replaced with new cables, some cables will be relabeled and reused, and new cables will be routed and connected. Refer to Section 4 cable maps for information on removing and installing cables

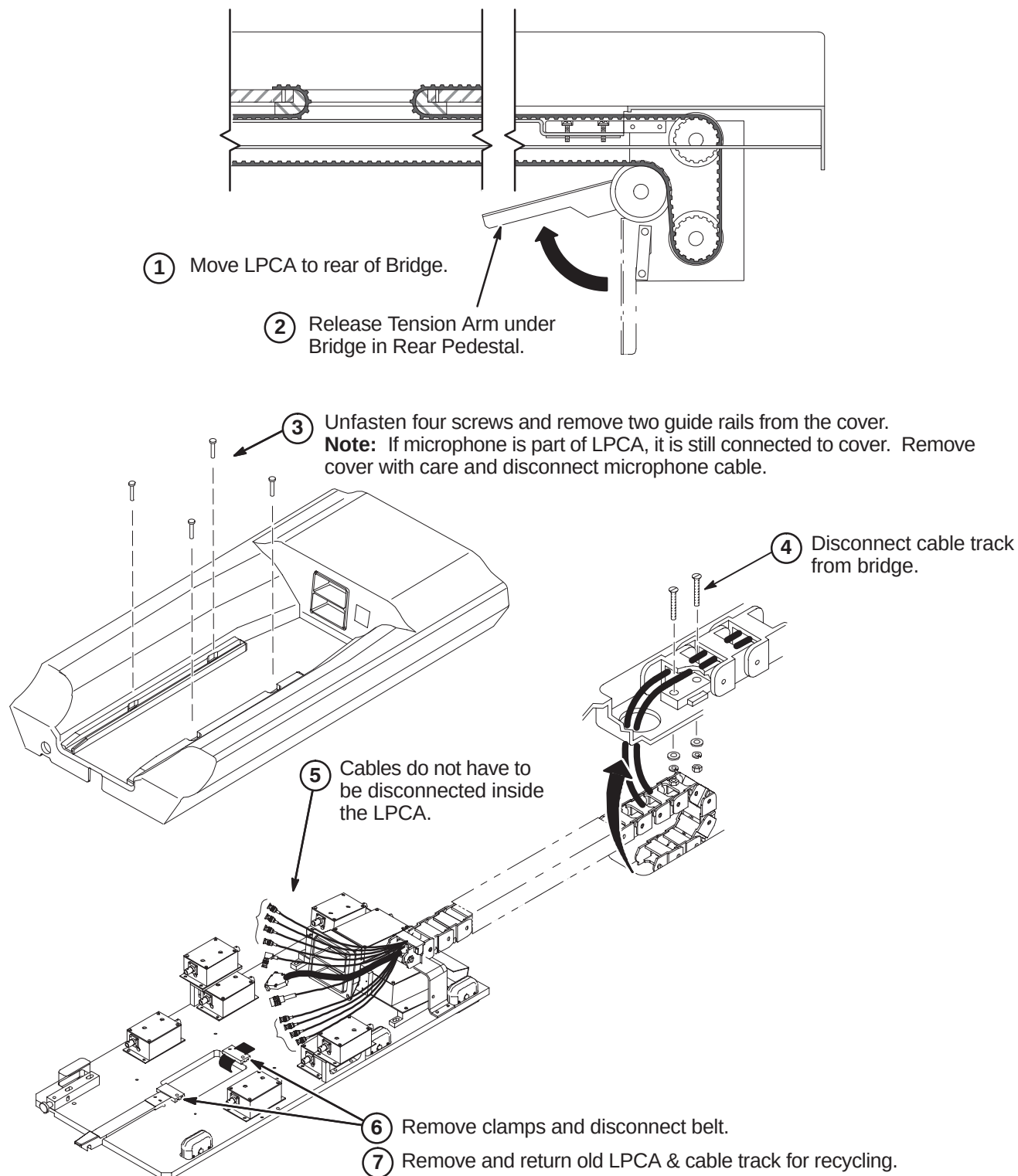
Also included in this section are the instructions for upgrading the Cradle Release on the Patient Table. Refer to section if the linkage mechanism does not have the same pieces as the new linkage shown in this procedure.

6-2 DEINSTALLATION PROCEDURES**6-2-1 DISCONNECT CABLE TRACK CABLES FROM REAR PEDESTAL MODULES****Note:**

Refer to cable removal map for information on any other cables that need to be disconnected and/or removed.

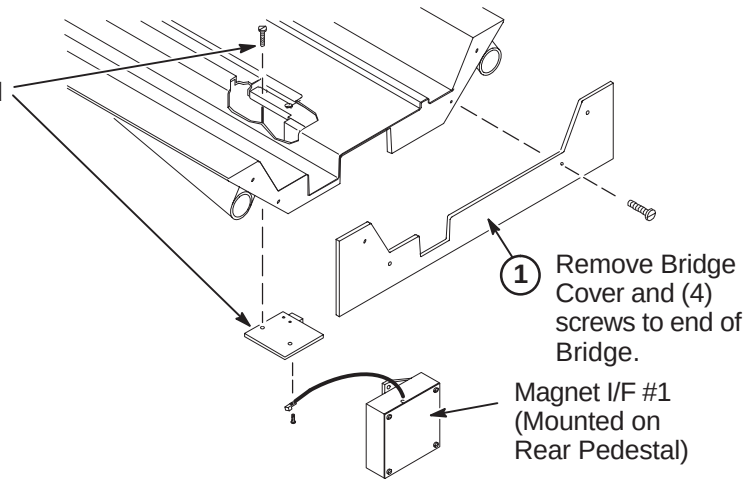
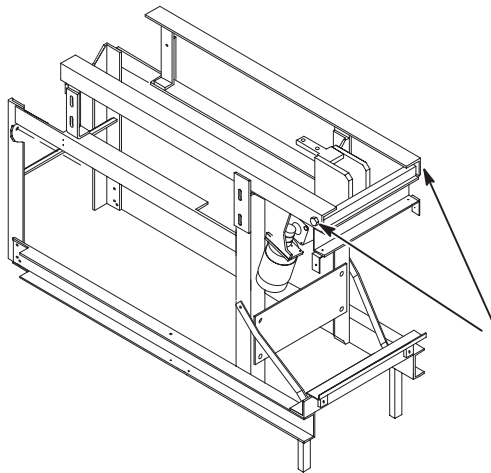


6-2-2 DEINSTALL EXISTING LPCA & CABLE TRACK



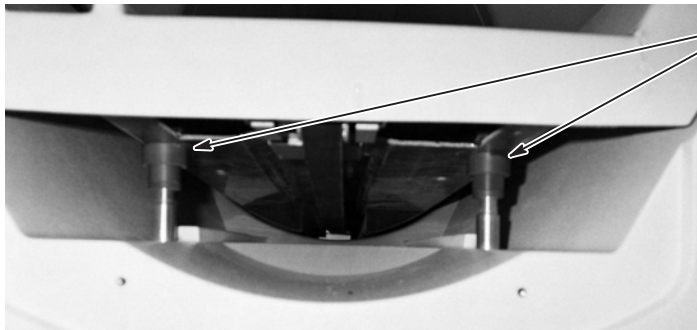
6-2-3 DEINSTALL BRIDGE**6-2-3-1 Unfasten Bridge from Rear Pedestal**

- ② Unfasten and save two 10-32 x 1-1/2 Round Head Slot Brass Screws (46-220184P47) and disconnect Bridge from existing Tongue and Rear Bridge Mount Asm.

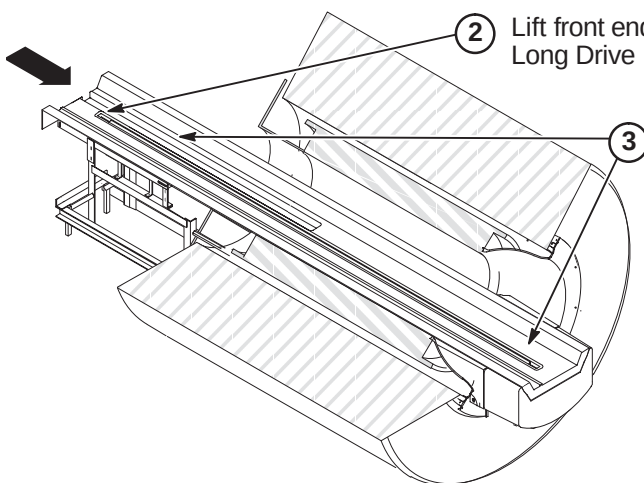


Note: If necessary, loosen, but do not remove, the screw on each side of the Longitudinal Drive Motor Asm so the Tongue will tilt up for easier removal of Bridge.

CAUTION: Longitudinal Motor contains ferrous material. Holding this assembly too close to magnet bore will forcibly attract it to magnet and possibly cause bodily injury or material damage.

6-2-3-2 Remove Bridge from Magnet

- ① Under the front of the Bridge, two Cap Support Cups (2318948) are epoxied to the bottom side of the Bridge. Remove these from the existing de-installed Bridge. If new Cups are not provided with upgrade, these will be reused on new Bridge.



- ② Lift front end of Bridge to disengage tongue on Long Drive assembly at Rear Pedestal.

- ③ While one person holds the front end, the other person stands in back of the bore next to the bridge and guides the Bridge Asm through the bore towards the front as it is being removed.

- ④ Remove and save Drive Belt and Front Pulley for transfer to new Bridge.

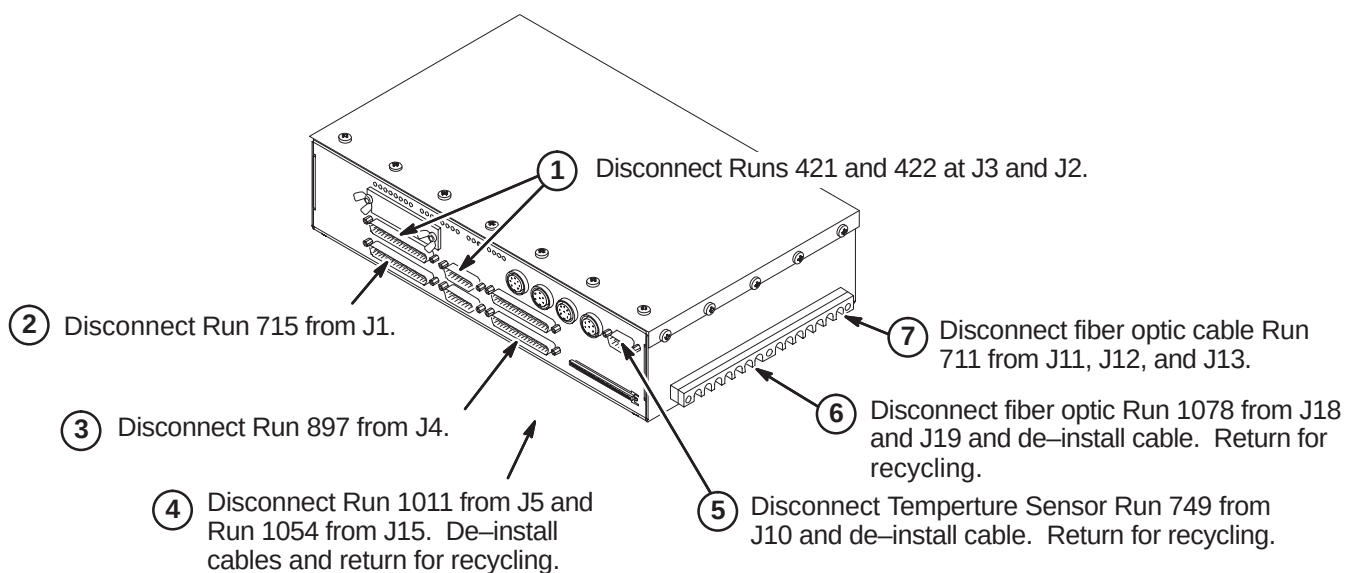
6-2-4 DEINSTALL SRI MODULE

The existing SRI Module will be replaced by the SRI-3 Module.

NOTE: Existing Runs 749 and 1011 must be removed and replaced with new Runs 1260 and 1261.

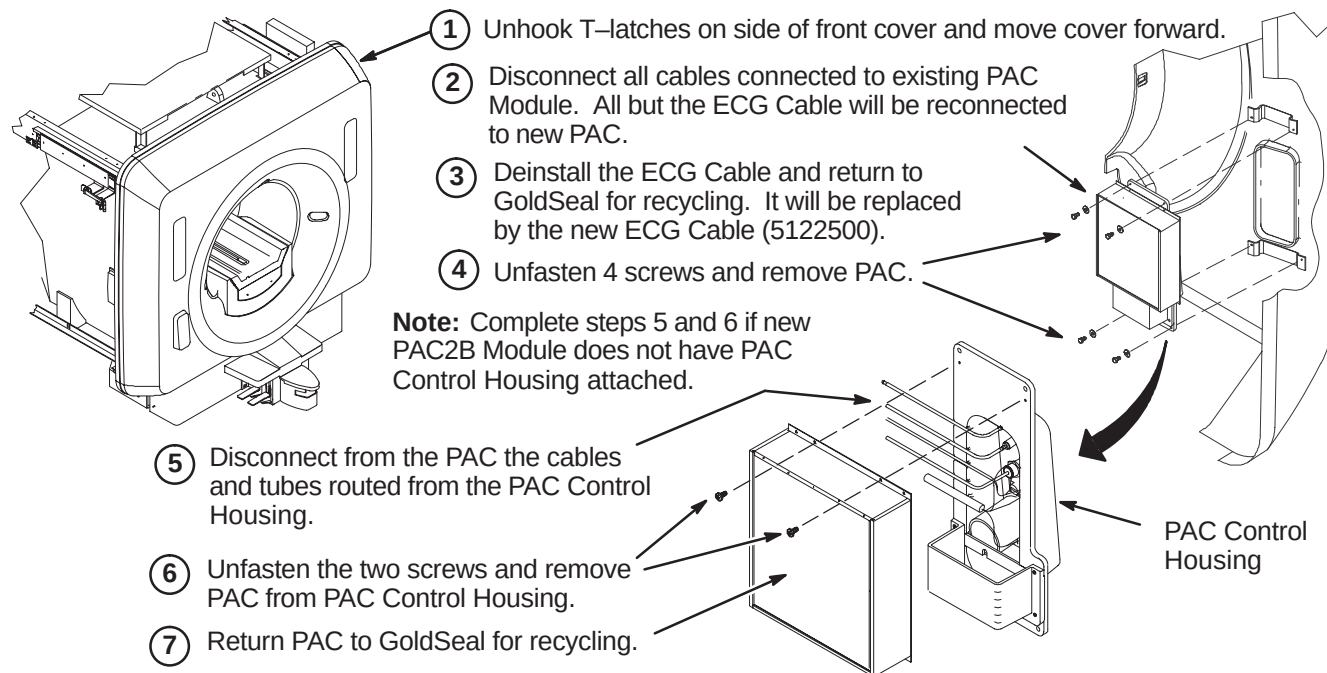


⑧ Carefully pull SRI Module off of velcro strips and return for recycling.

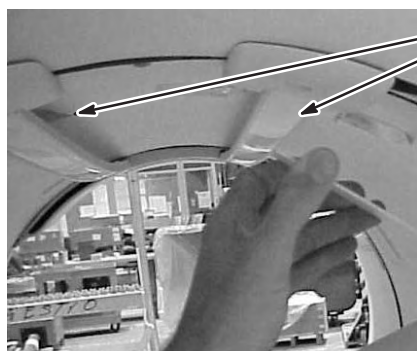
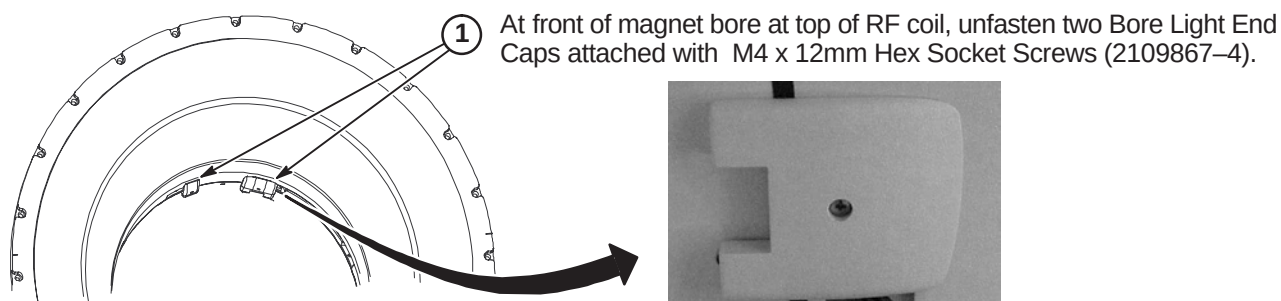


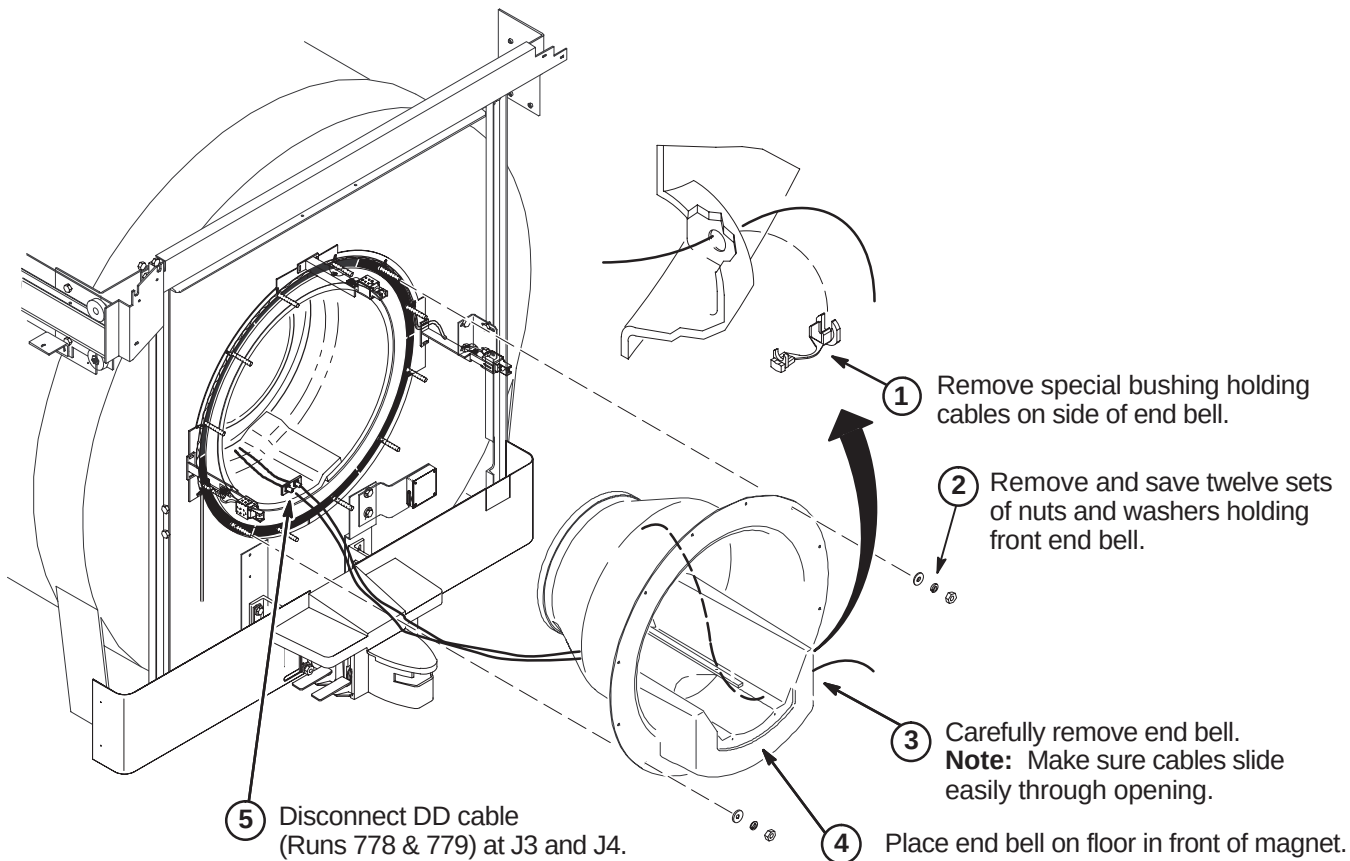
6-2-5 DEINSTALL PAC MODULE

The existing PAC Module will be replaced by a new PAC 2B Module with independent vector gating.

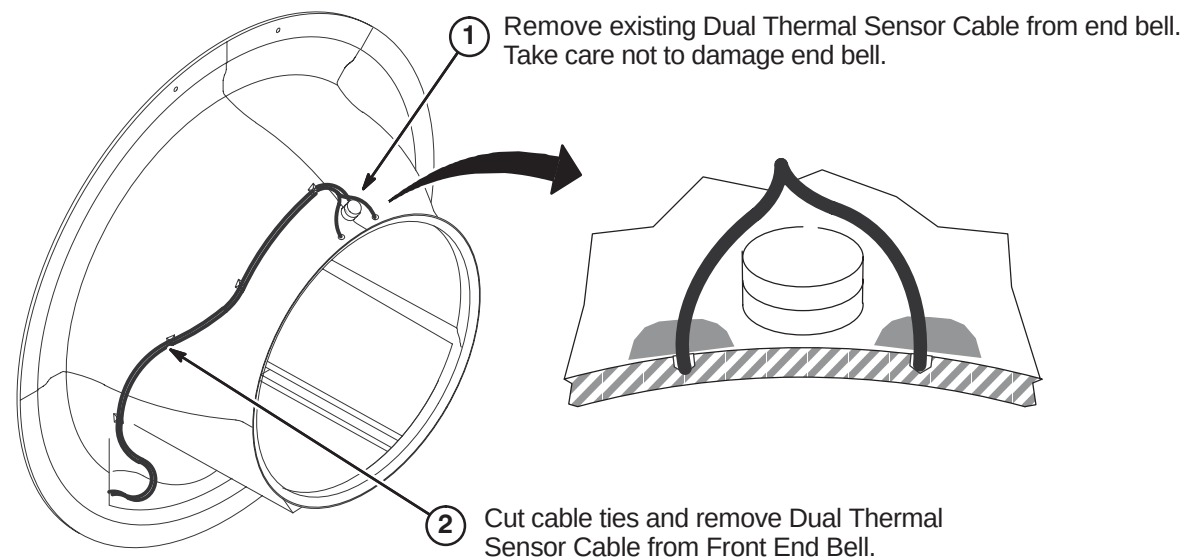
**6-3 INSTALL NEW DUAL TEMPERATURE SENSOR CABLE ON FRONT END BELL**

The existing Temperature Sensor Cable, Run 749, needs to be removed and must be replaced with new Dual Thermal Sensor cable, Run 1260.

6-3-1 UNFASTEN FLAT PANEL BORE LIGHTS

6-3-2 REMOVE FRONT END BELL**6-3-3 DE-INSTALL EXISTING RUN 749**

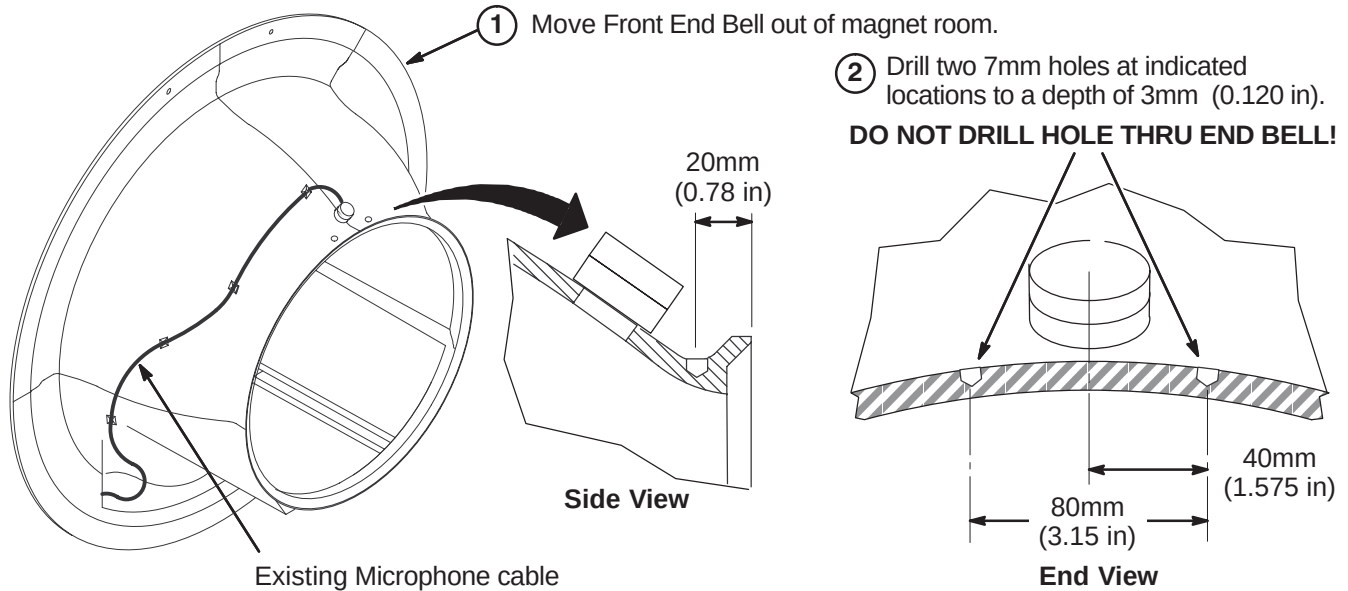
The site may have upgraded to a Dual Thermal Sensor Run 749 recently, but this cable has to be removed and replaced with new Dual Thermal Sensor cable, Run 1260 (5123812). EXCITE HD requires Run 1260 to be installed.



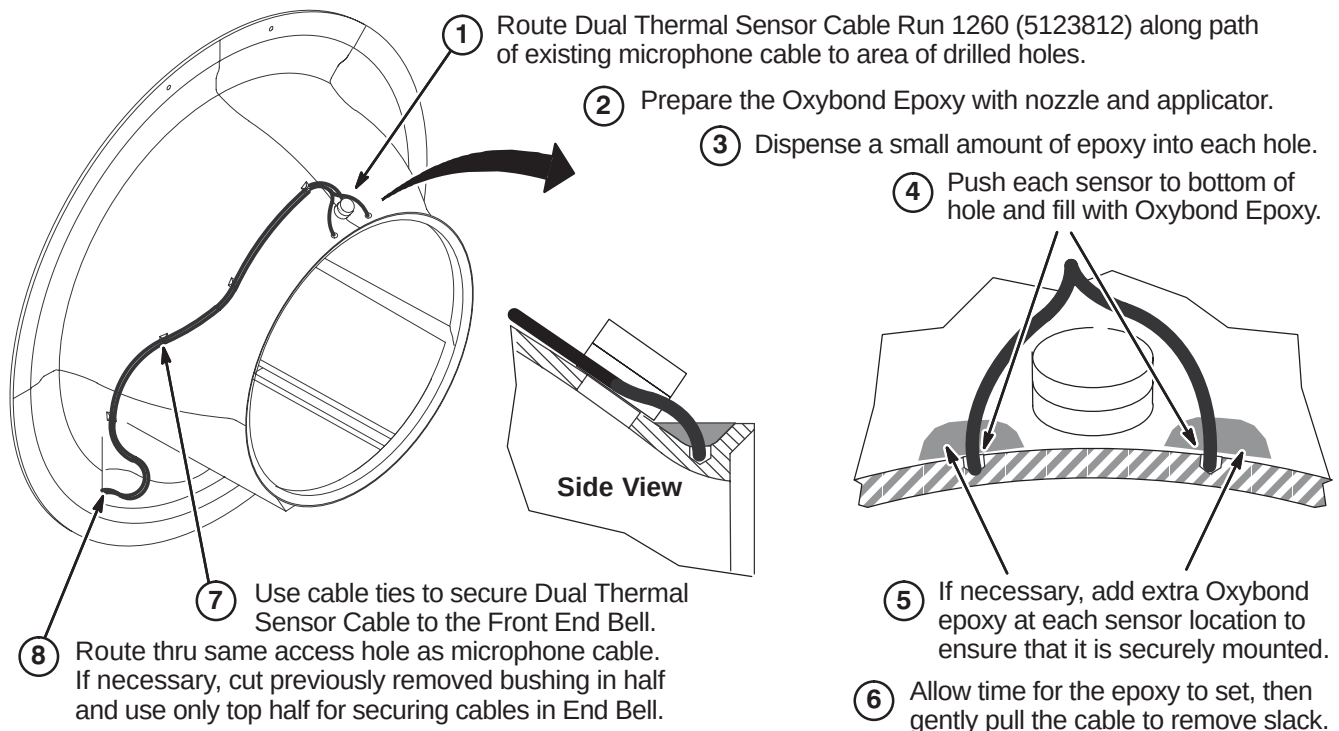
6-3-4 PREPARE END BELL FOR INSTALLATION OF DUAL THERMAL SENSOR CABLE

WARNING!

FERROUS MATERIAL HAZARD! DO NOT DRILL, HAMMER, CHISEL, ETC. OR USE ANY FERROUS TOOLS IN MAGNET ROOM UNLESS MAGNET IS RAMPED DOWN!

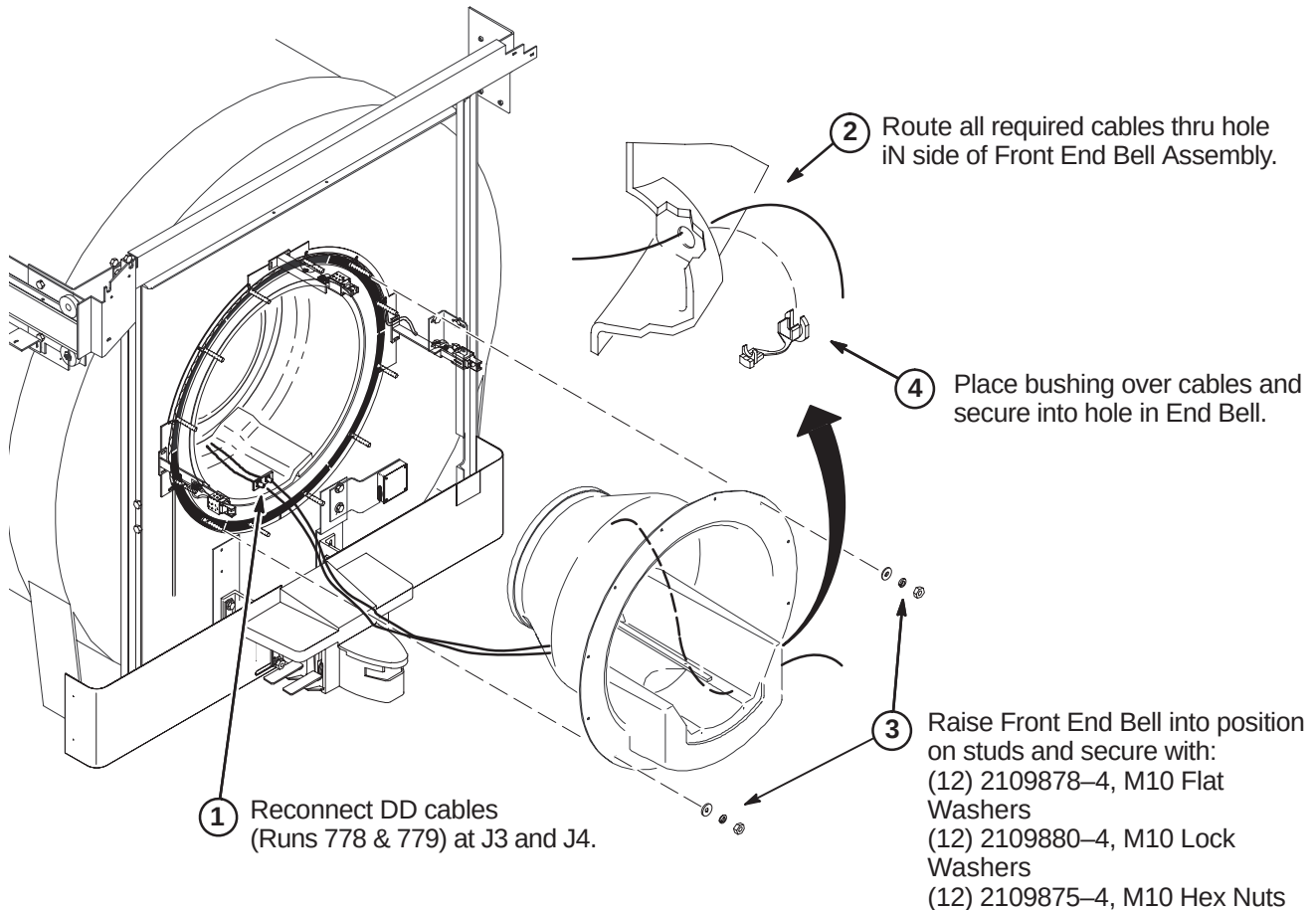
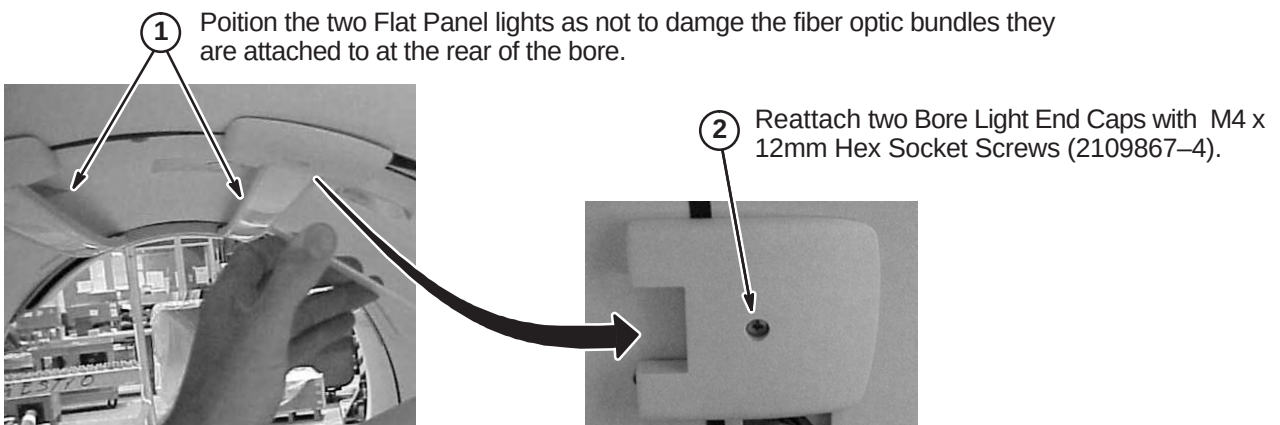


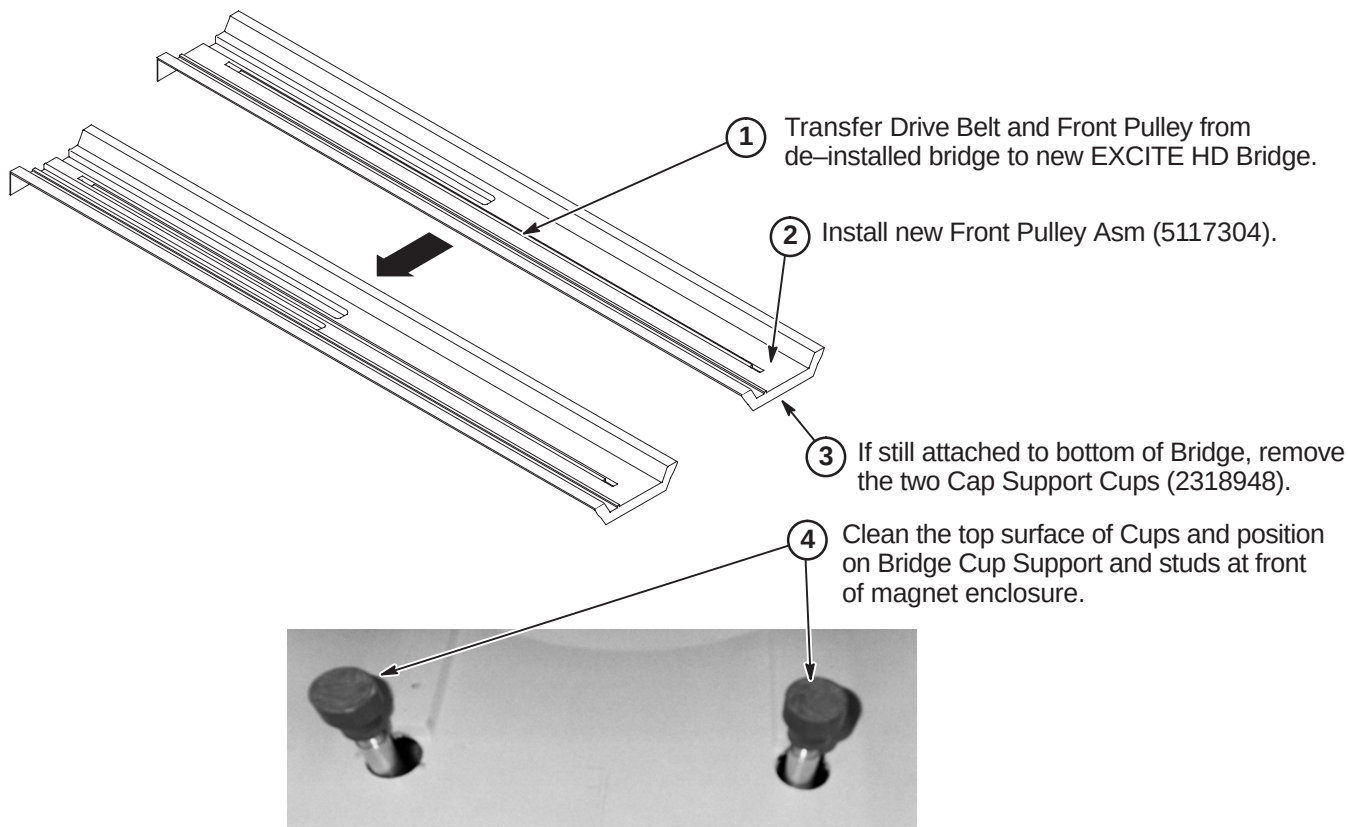
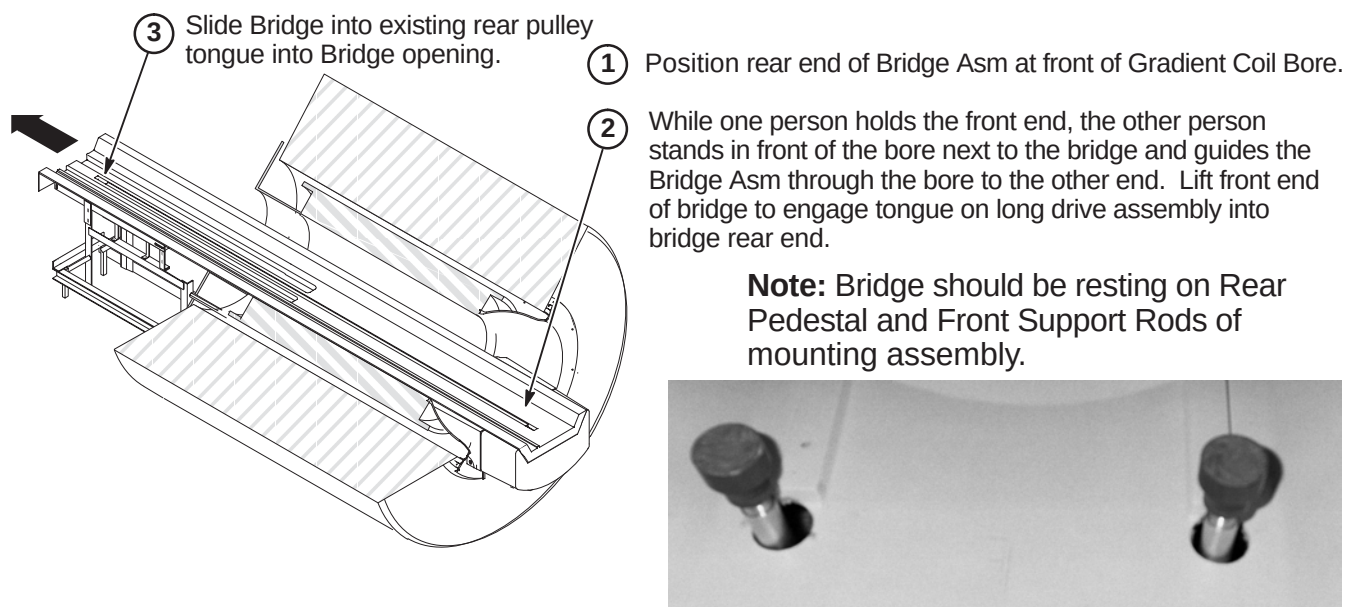
6-3-5 ATTACH DUAL THERMAL SENSOR CABLE

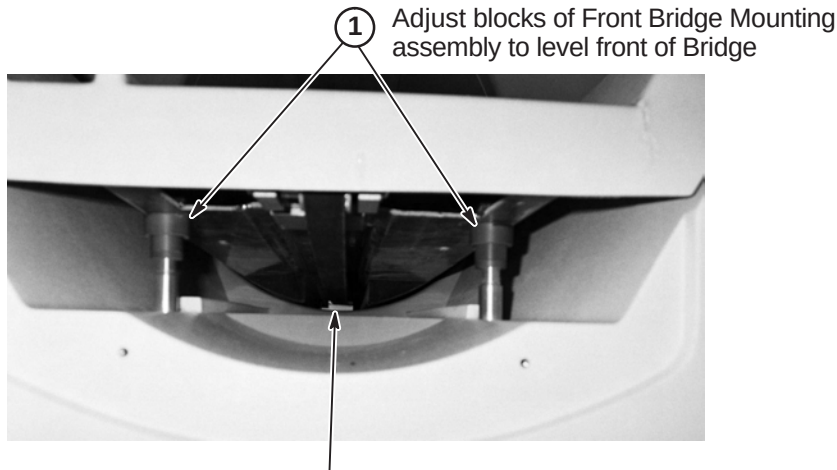


6-3-6 ATTACH FRONT END BELL TO MAGNET

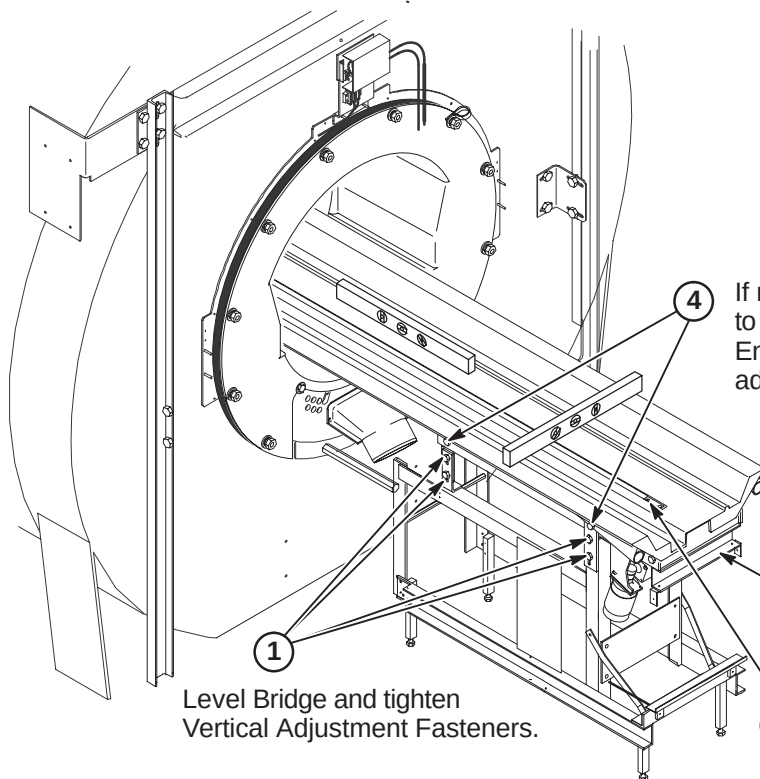
Note: A good seal is required to prevent air leaks. Any air leaks between RF Tube (Gradient Coil) and End Bell joint can cause a loud humming noise during patient scanning. If possible, conduct an air leakage test prior to Bridge installation. This would require connecting power to the Blower Box and connecting 6" blower hose from Blower Box to Bulkhead.

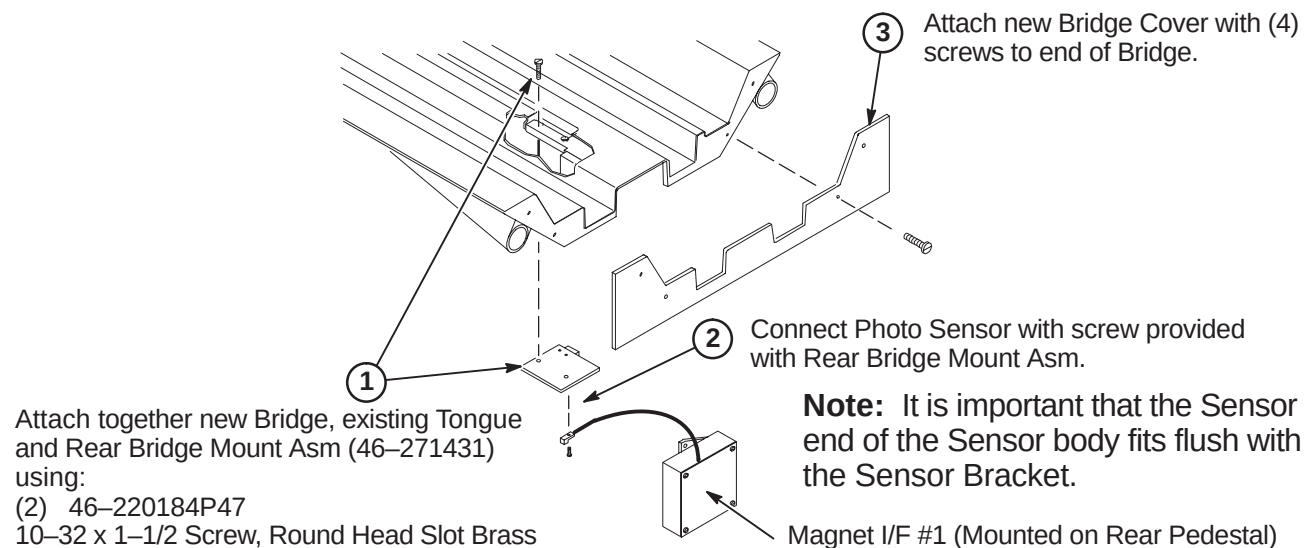
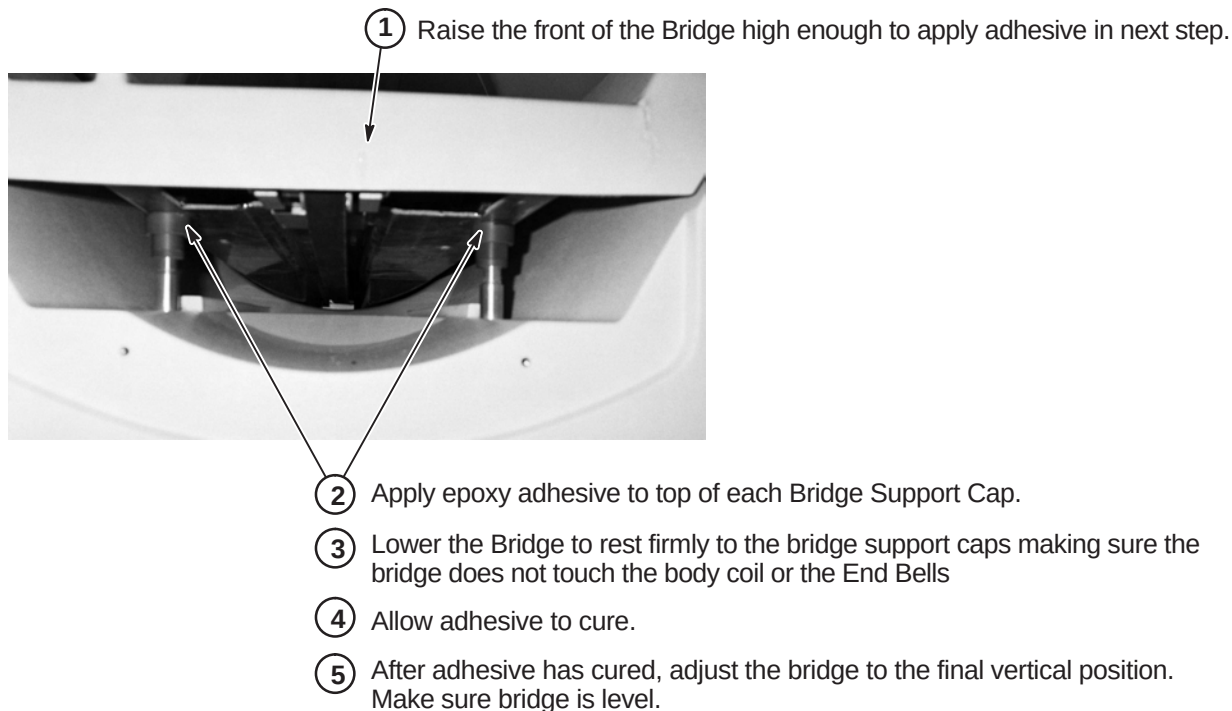
**6-3-7 ATTACH BORE LIGHTS**

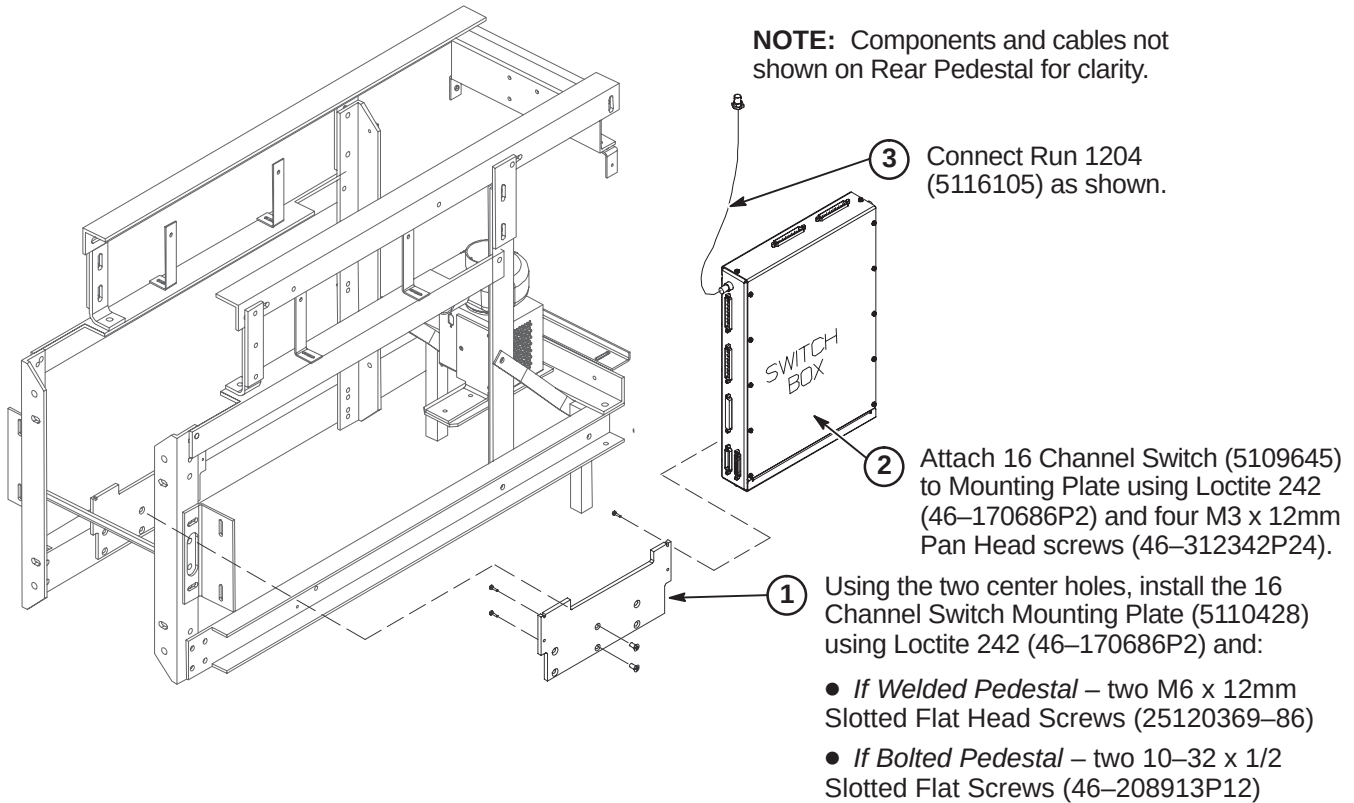
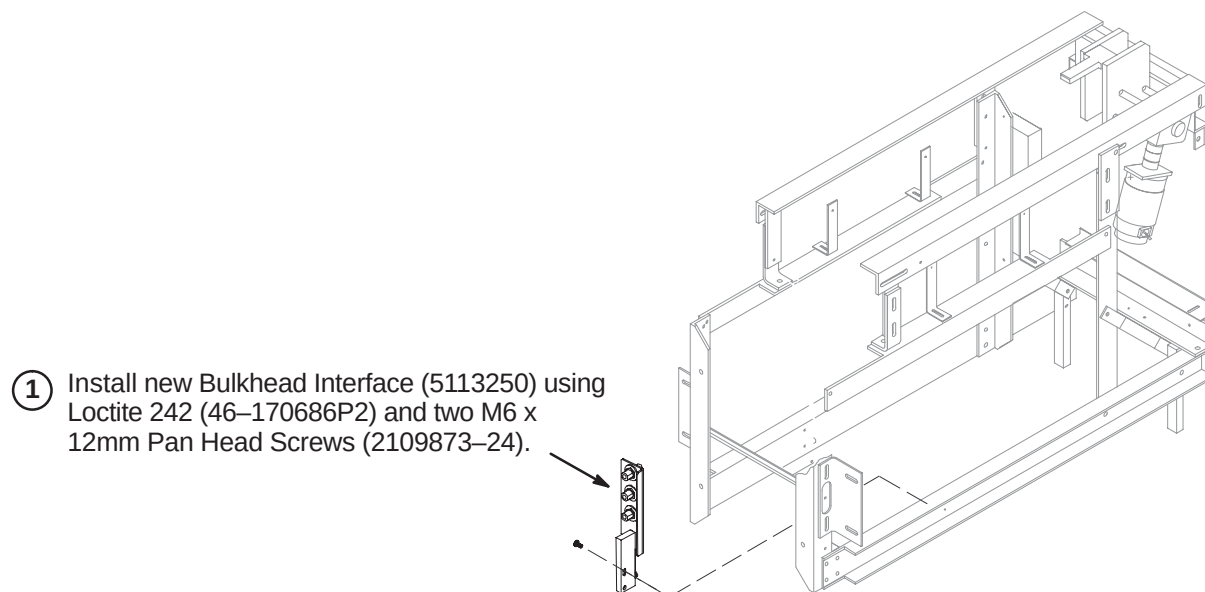
6-4 INSTALL NEW COMPONENTS AND CABLES**6-4-1 INSTALL NEW DUAL TRACK BRIDGE****6-4-1-1 Transfer Components to New Bridge****6-4-1-2 Install Bridge**

6-4-1-3 Level Front of Bridge**IMPORTANT!**

THE BRIDGE MUST NOT COME IN CONTACT WITH THE RF COIL.
There must be an air gap of 6mm to 8mm thru entire length of bore.

6-4-1-4 Level All of Bridge

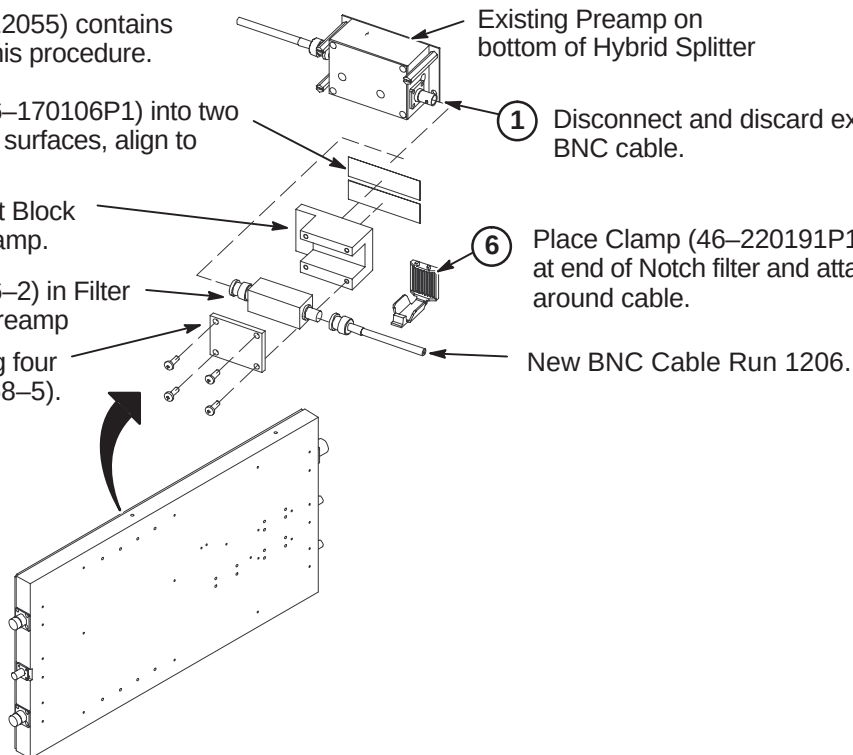
6-4-1-5 Connect Photo Sensor and Bridge End Cover**6-4-1-6 Apply Adhesive to Front Bridge Support Caps**

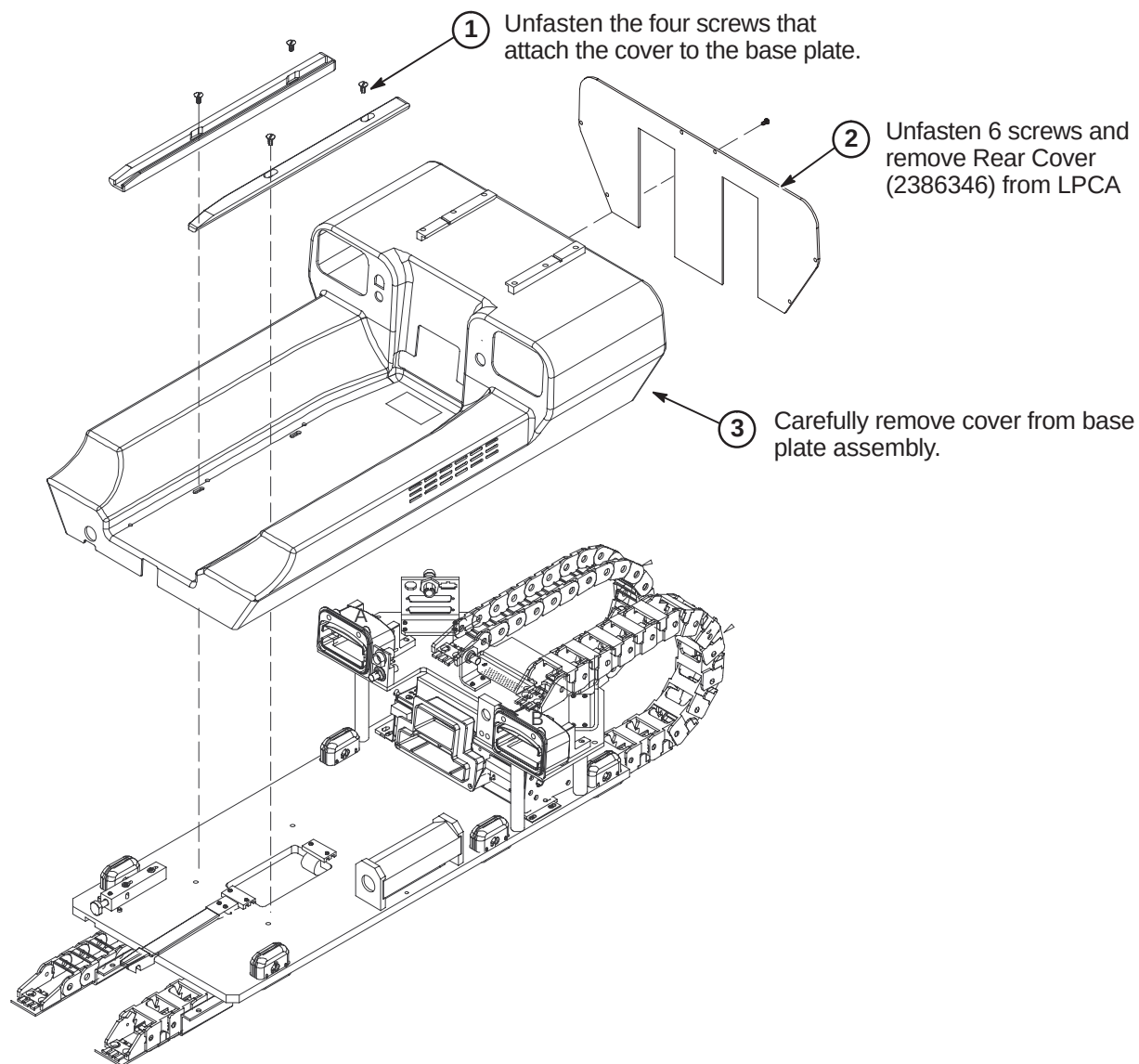
6-4-2 INSTALL NEW COMPONENTS ON REAR PEDESTAL**6-4-2-1 Install 16 Channel Switch****6-4-2-2 Install Bulkhead Interface**

6-4-2-3 Install Filter Mount Asm and Isolation Filters to Body Hybrid Splitter

NOTE: The Filter Mount Asm (5112055) contains several parts that are installed in this procedure.

- ① Disconnect and discard existing BNC cable.
- ② Cut Double Coated Film (46-170106P1) into two strips 70mm long. Clean all surfaces, align to Preamp and attach.
- ③ Align and attach Filter Mount Block (5112053) with existing Preamp.
- ④ Position Notch Filter (2305086-2) in Filter Mount Block and connect to Preamp
- ⑤ Attach Filter Clamp Plate using four M4 x 12 Nylon screws (2112068-5).
- ⑥ Place Clamp (46-220191P1) at end of Notch filter and attach around cable.

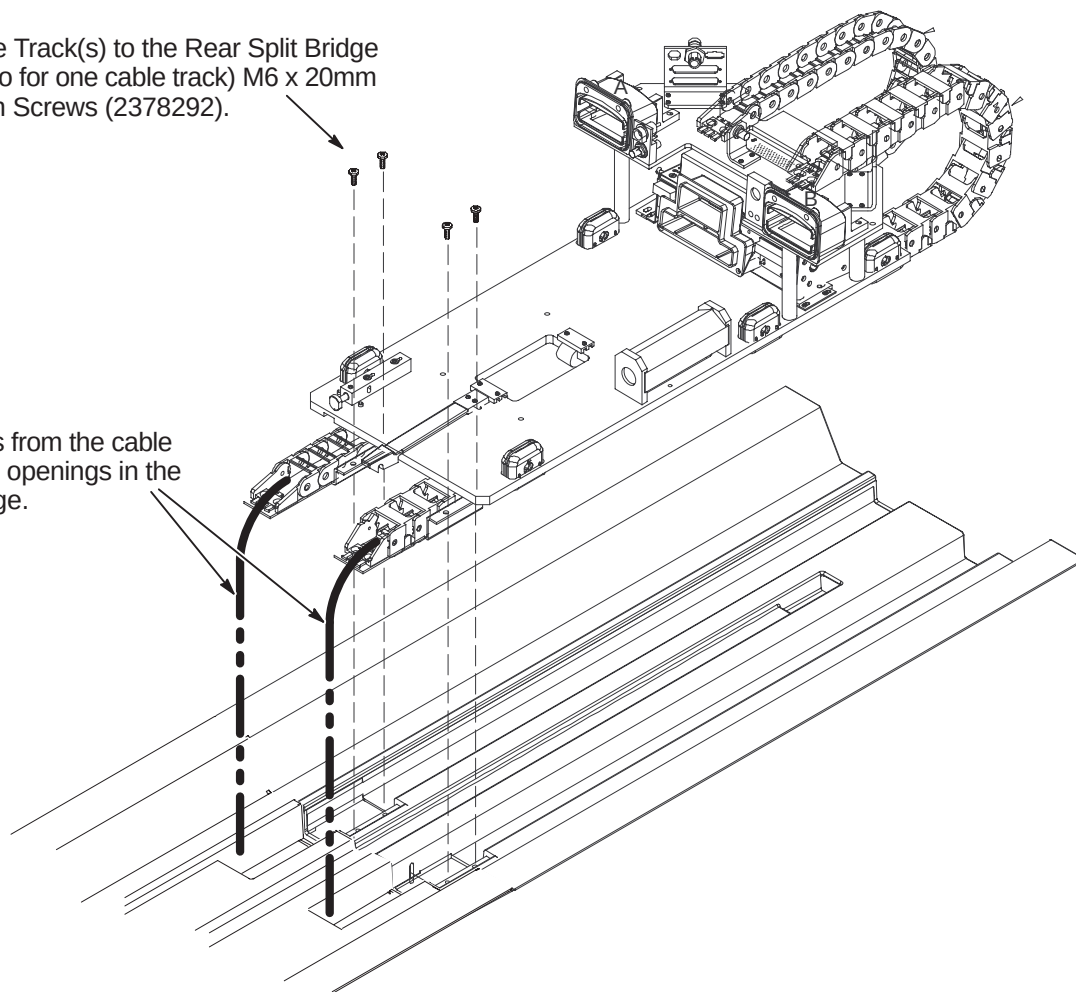


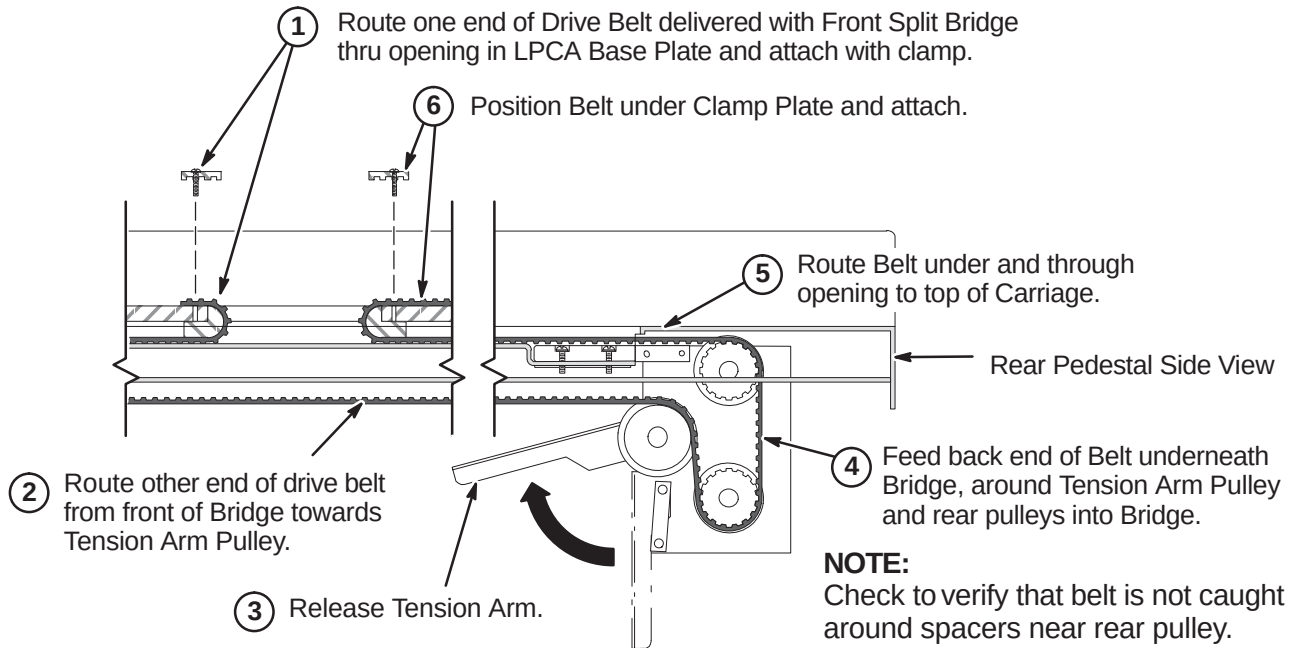
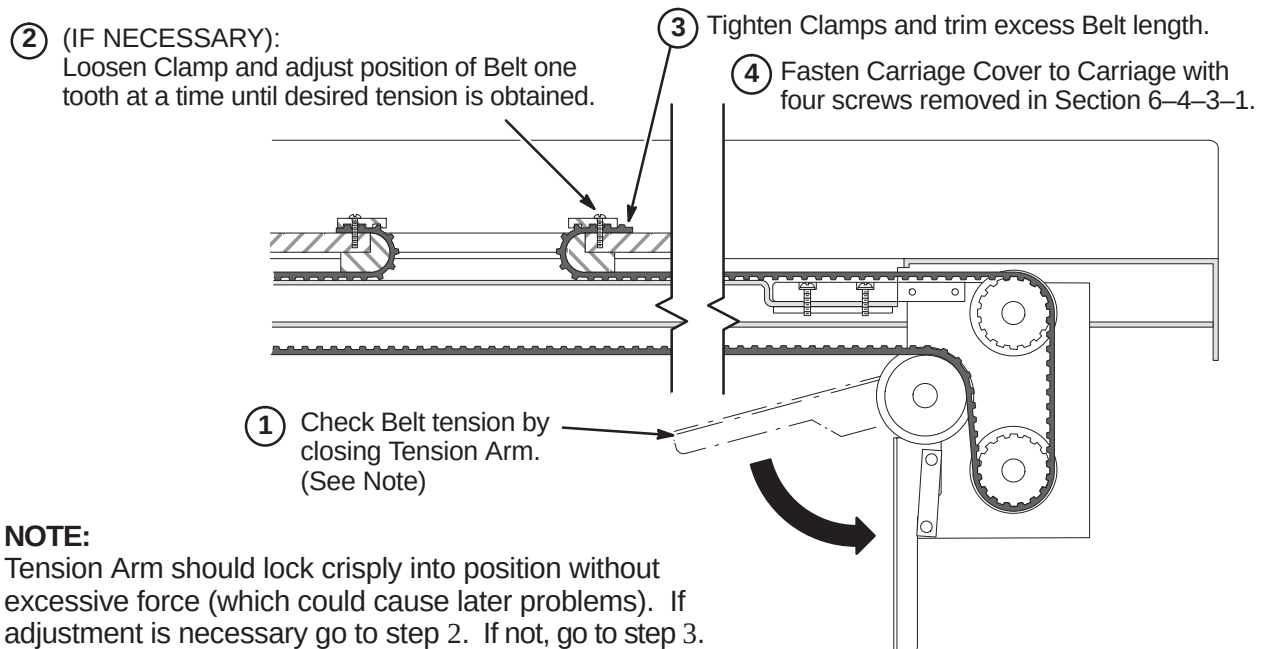
6-4-3 INSTALL LPCA AND CABLE TRACKS**6-4-3-1 Remove LPCA Cover**

6-4-3-2 Attach Cable Tracks to Bridge

- ② Attach the Cable Track(s) to the Rear Split Bridge using four (or two for one cable track) M6 x 20mm Pan Head Nylon Screws (2378292).

- ① Feed the cables from the cable track(s) thru the openings in the Front Split Bridge.



6-4-3-3 Route and Attach Drive Belt**6-4-3-4 Final Adjustment of Drive Belt**

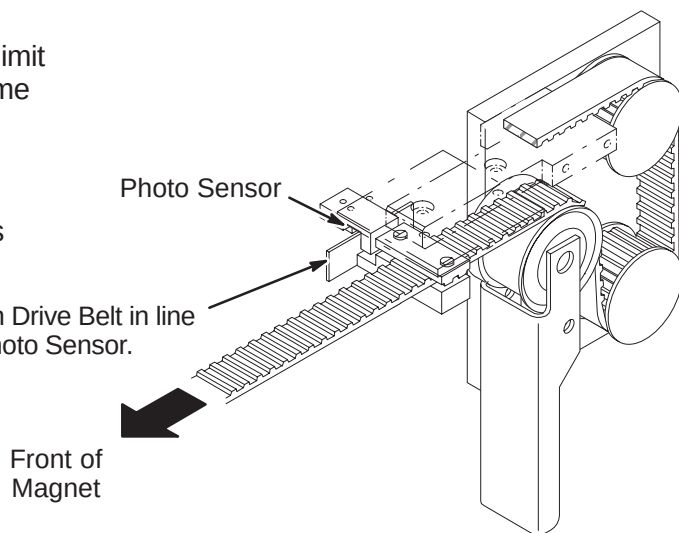
6-4-3-5 Installation of Sensor Flag**NOTE:**

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

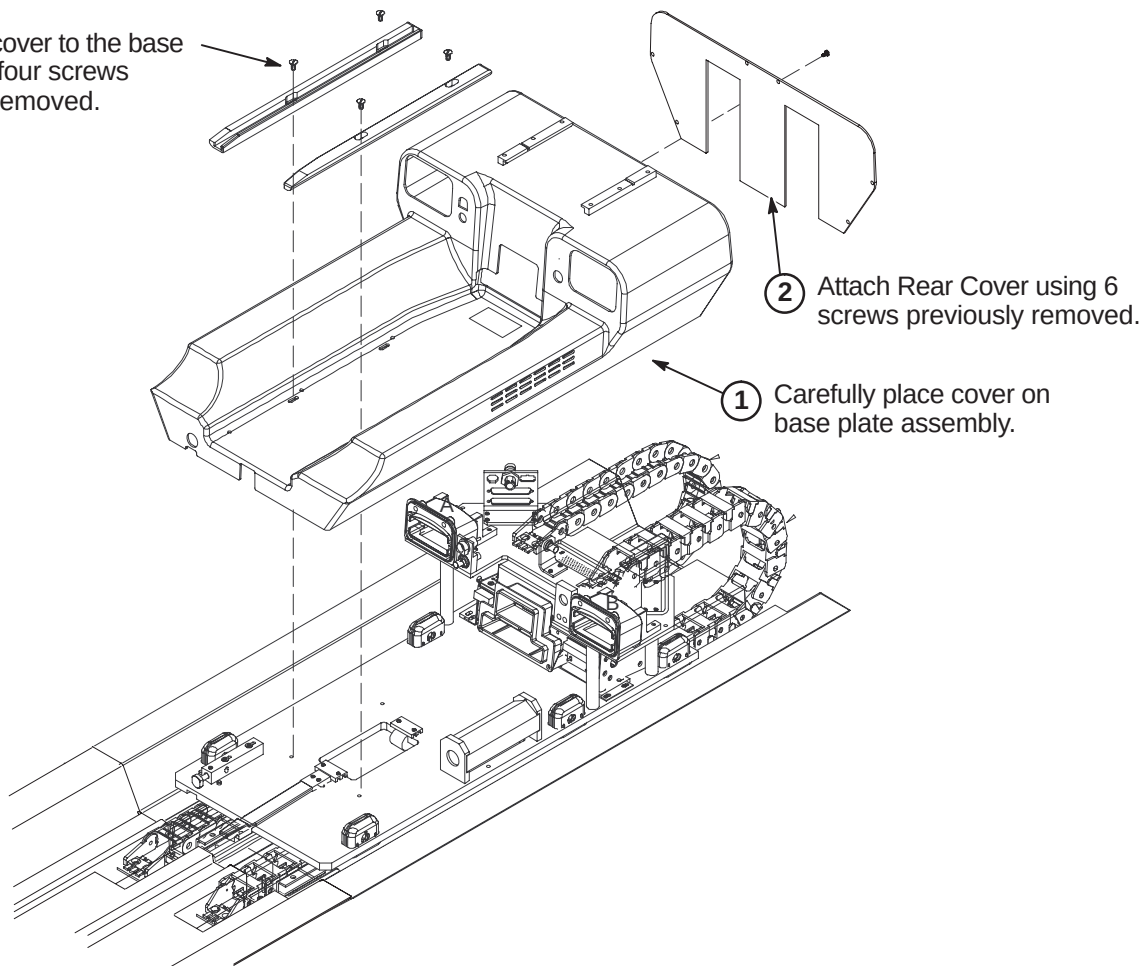
NOTE:

The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.

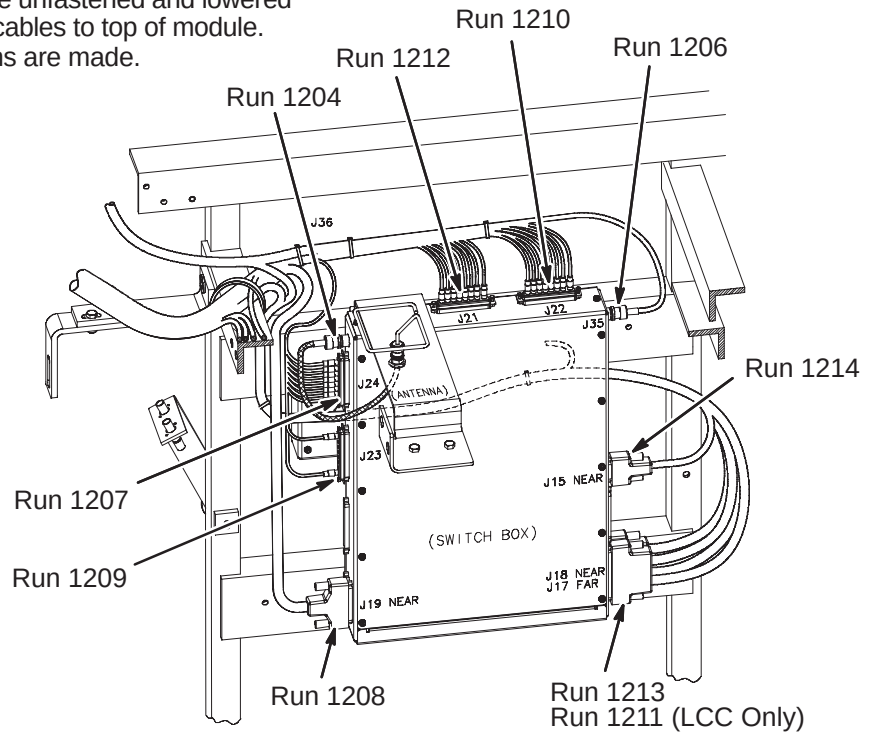
**6-4-3-6 Installation of LPCA Cover**

- ③ Attach the cover to the base plate using four screws previously removed.



6-4-3-7 Connect Cables from Cable Track to 16 Channel Switch Module

- ① 16 Channel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re-mount module after connections are made.
- ② Connect cable Runs as indicated.

**6-4-3-8 Connect Cables from Cable Track to Bulkhead Interface**

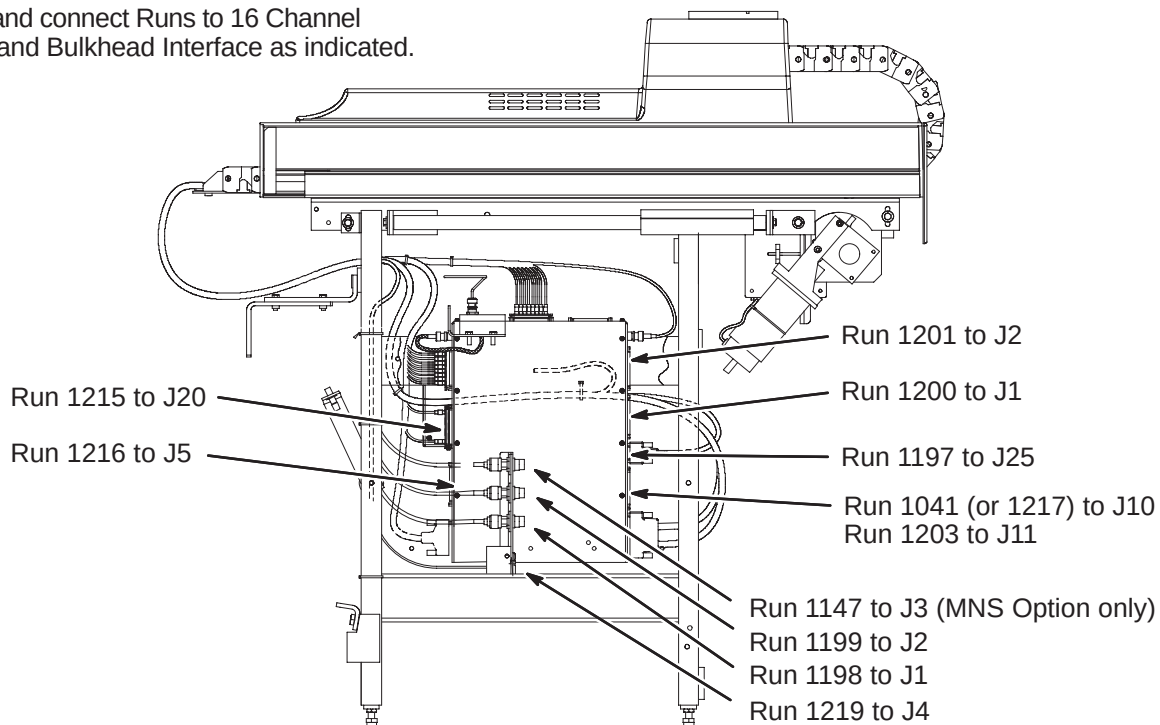
- ① Route Runs 1153, 1205, 1206, and 1228 along leg of Rear Pedestal and ty-wrap as indicated to leg.

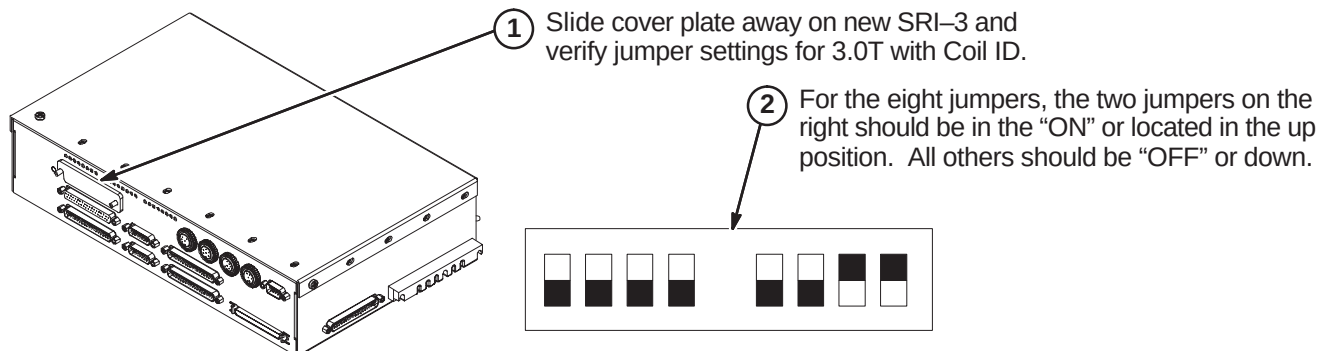
- ② Connect Runs to Bulkhead Interface as indicated.

Run 1153 to J3
Run 1205 to J2
Run 1228 to J1
Run 1206 to J4

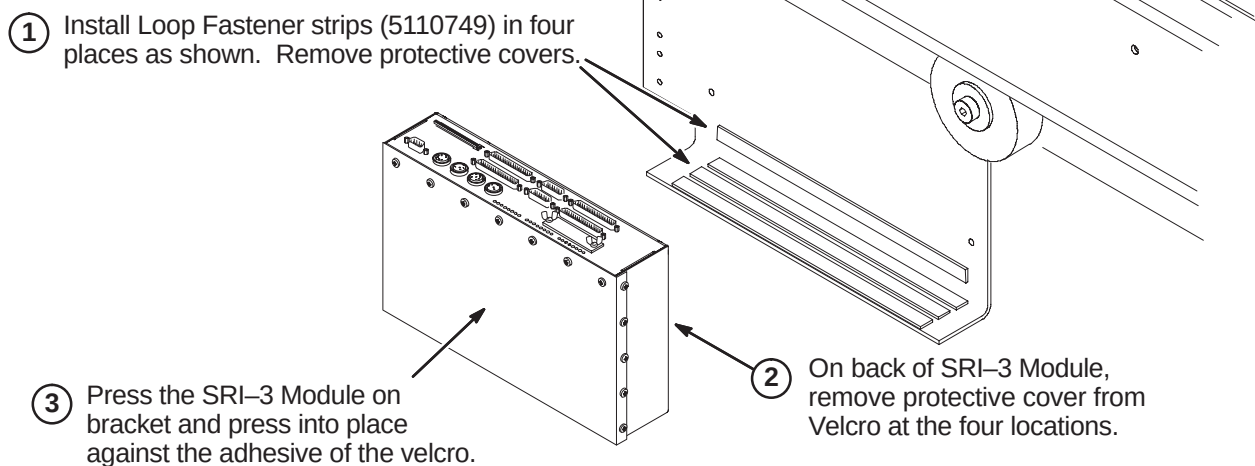
6-4-4 CONNECT REMAINING CABLES TO REAR PEDESTAL**6-4-4-1 Cable Connections to 16 Channel Switch and Bulkhead Interface**

- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.

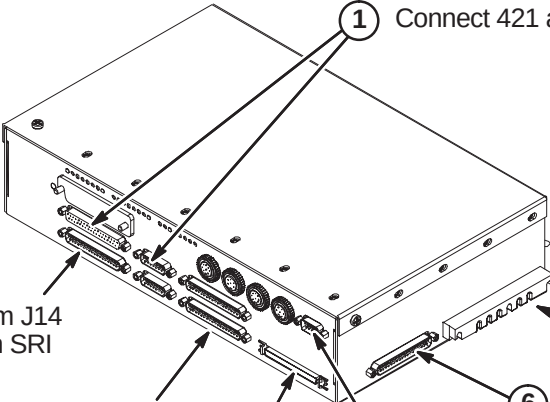


6-4-5 INSTALL SRI-3 MODULE**6-4-5-1 Verify Jumper Positions for SRI-3****6-4-5-2 Position SRI-3 in L Bracket****Note:**

SRI-3 Module to be installed so that it can be easily pulled off for the installation and removal of cables.



6-4-5-3 Route and Connect Cables to SRI-3

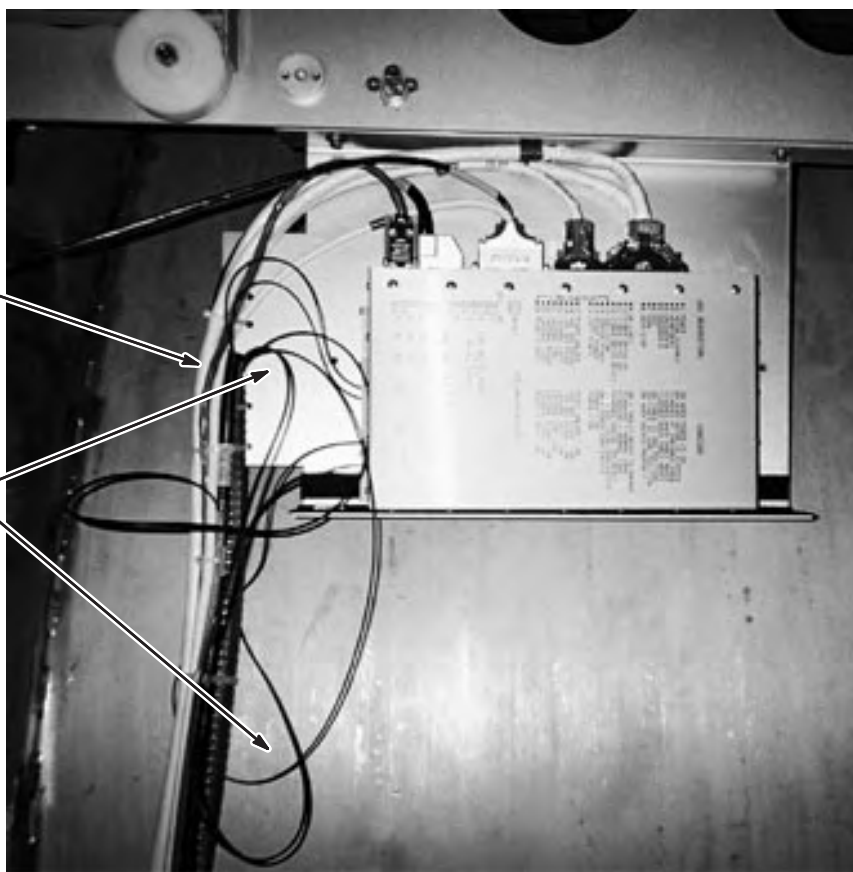
- 
- ① Connect 421 and 422 to J3 and J2.
- ② Connect Run 715 from J14 on Pen Panel to J1 on SRI Module.
- ③ Connect Run 897 from Operator Display to J4.
- ④ Route Run 1261 from Operator Display and connect to J5.
- ⑤ Route Run 1260, Dual Temperature Sensor, from Front End Bell and connect to J10.
- ⑥ Route Run 1215 from J20 on 16 Channel Switch and connect to J20.
- ⑦ Connect fiber optic cable Run 711 to J11, J12, and J13.

- ⑧ Route all cables and fiber optic cables as shown.

Note:

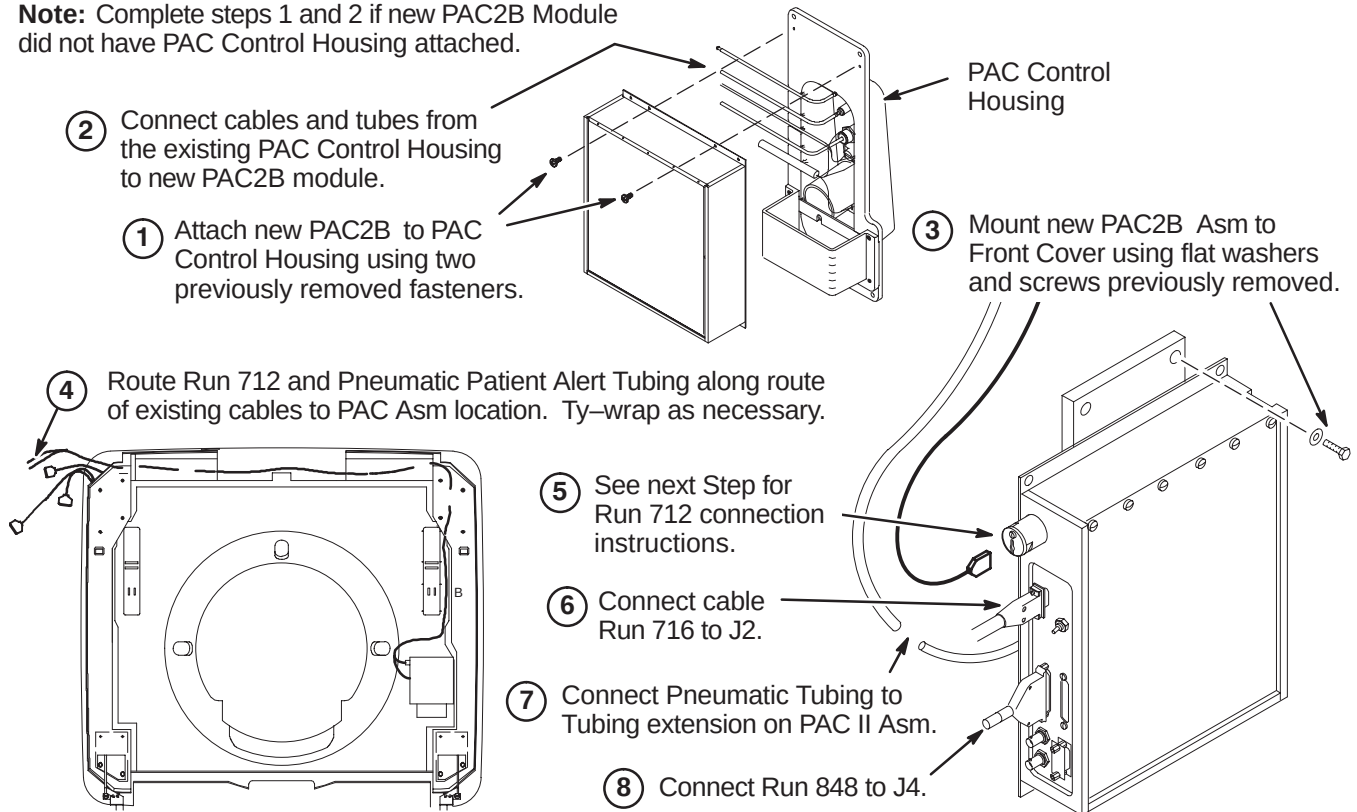
Fiber optic cables loops must not be smaller than two inch (50mm) radius.

- ⑨ Reapply spiral wrap around all cables and fiber optics.



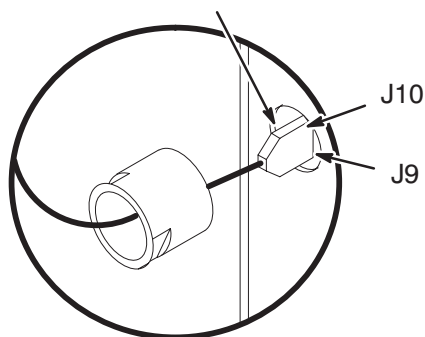
6-4-6 INSTALL PAC MODULE**6-4-6-1 Connect Run 712, 716, & Pneumatic Patient Alert Tubing**

Note: Complete steps 1 and 2 if new PAC2B Module did not have PAC Control Housing attached.

**6-4-6-2 Connect Run 712**

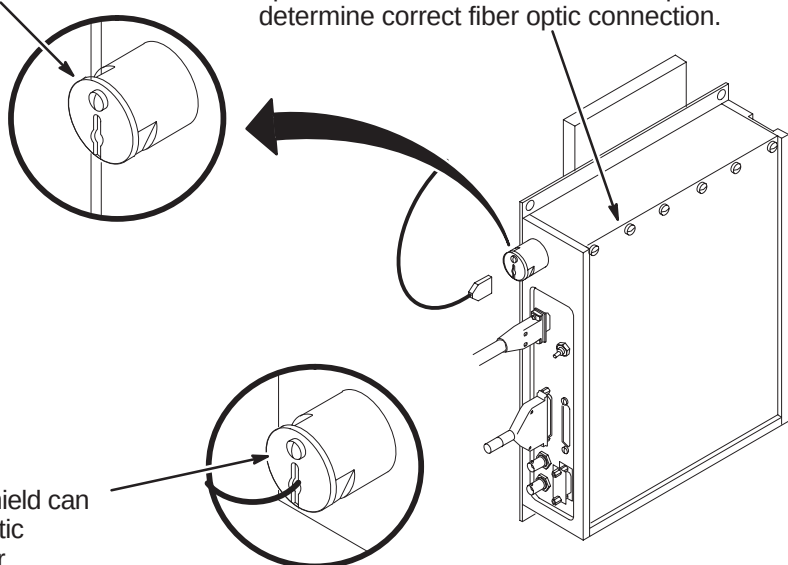
① Remove shield cover by removing one screw and unscrew shield from module.

② Route Run 712 fiber optic cables through shield into PAC module and connect to J9 and J10 on circuit board.



③ Screw shield in place by hand. Shield can be rotated slightly to ease fiber optic connection. Reinstall shield cover.

Note: Loosen screws and slide module cover open if access to PAC-II board is required to determine correct fiber optic connection.



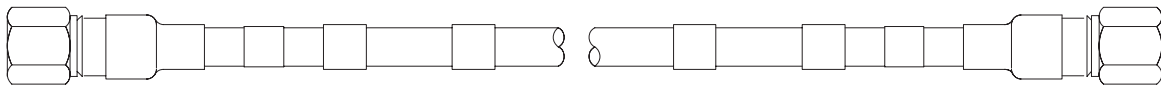
6-5 TIMES MICROWAVE RF CABLES INSTALLATION**6-5-1 CABLE INFORMATION**

Two cables are to be created and other associated parts to be installed are provided for this procedure. Also provided is a Times Microwave stripping tool for the LMR 600 (2352193), and a Times Microwave cutting tool (5111565).

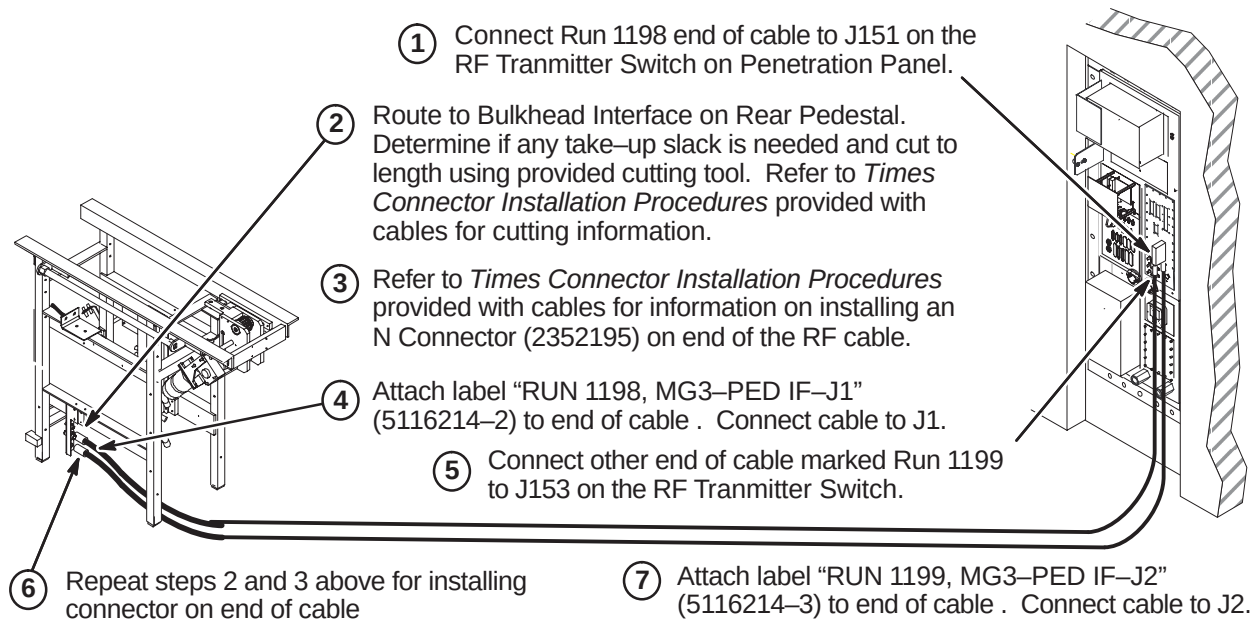


Do not subject LMR-600FR cable to a bend radius less than 3 inches (75mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

Cable 5115915, will be cut to length to create two cables, Runs 1198 and 1199, and terminated for routing in the scan room. The remaining connectors, fasteners, and labels will be installed as directed in the following steps.



Run 1198 and Run 1199

6-5-2 ROUTE, TERMINATE, AND CONNECT RF CABLES

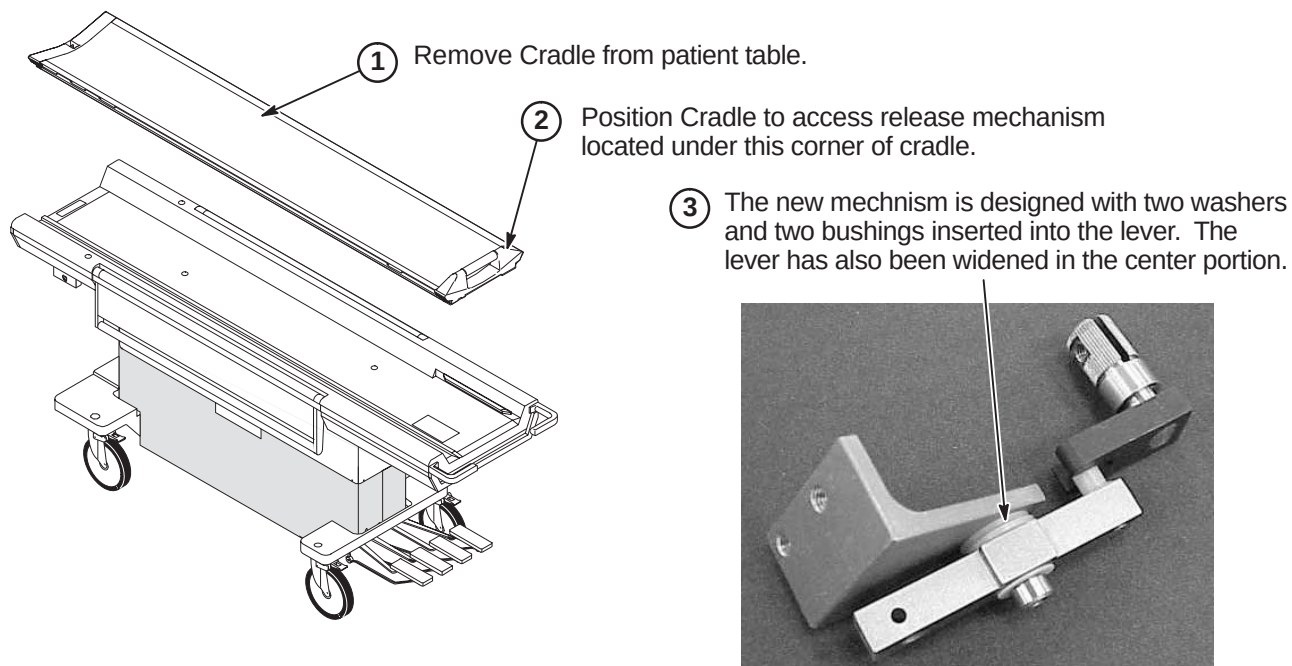
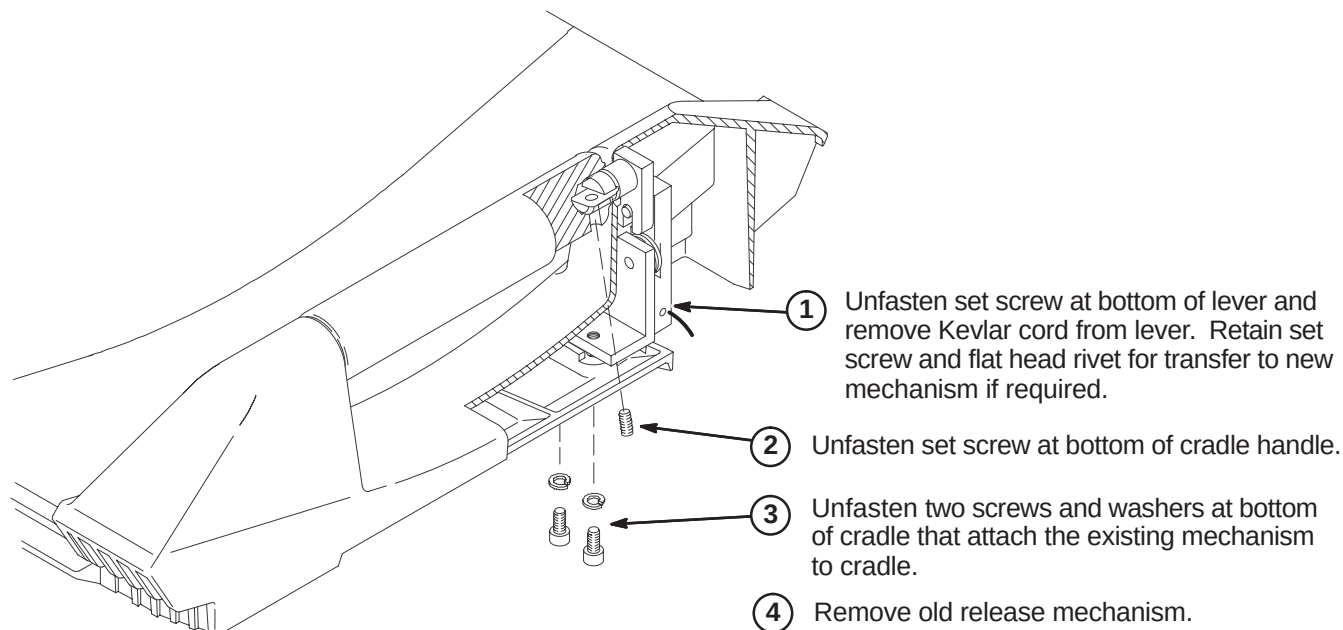
SECTION 7 – CRADLE RELEASE UPGRADE

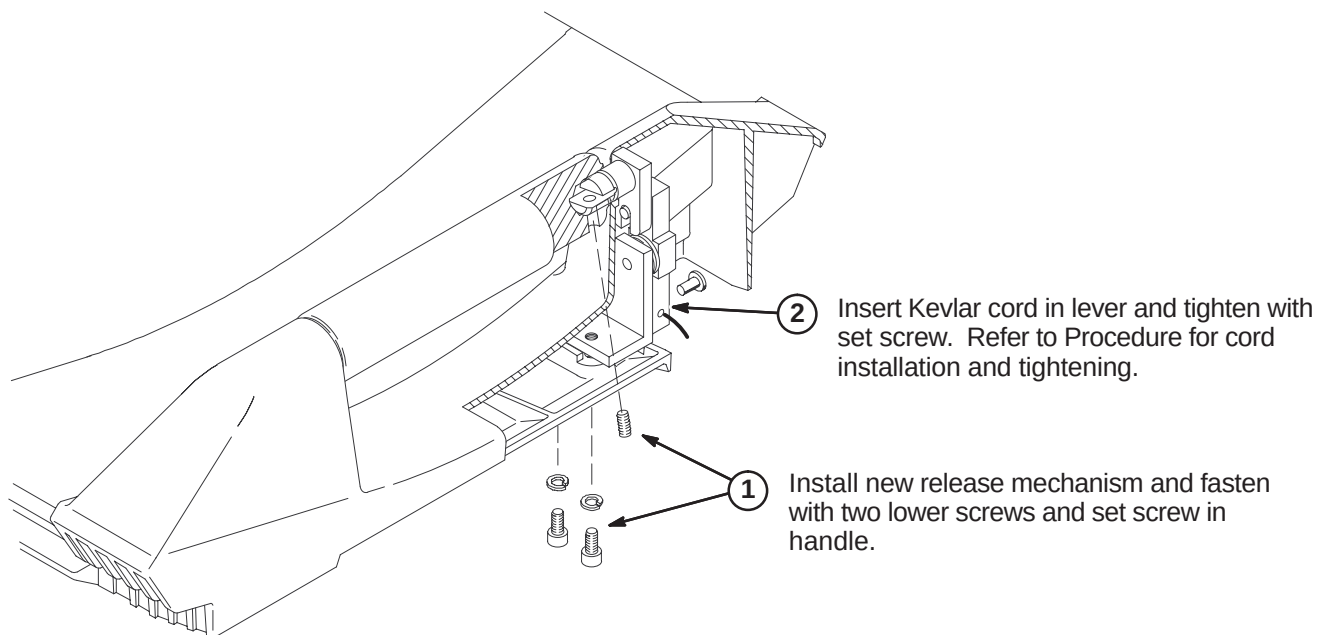
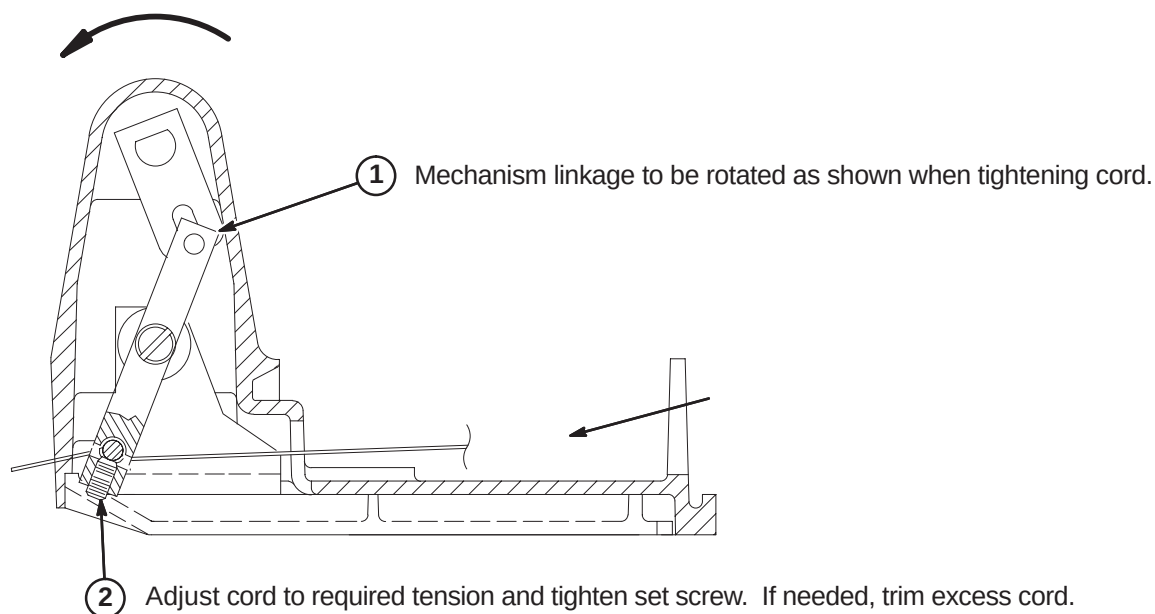
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<u>SECTION</u>	<u>PAGE</u>
7-1 PATIENT TABLE CRADLE RELEASE UPGRADE	7-2
7-1-1 PREPARE CRADLE FOR UPGRADE	7-2
7-1-2 UPGRADE PROCEDURE	7-2

7-1 PATIENT TABLE CRADLE RELEASE UPGRADE

This upgrade is performed to improve the release mechanism and improve image quality.

7-1-1 PREPARE CRADLE FOR UPGRADE**7-1-2 UPGRADE PROCEDURE****7-1-2-A Remove Existing Release Mechanism**

7-1-2-B Install New Release Mechanism**7-1-2-C Attach Cord**

SECTION 8 – OPERATOR WORKSPACE

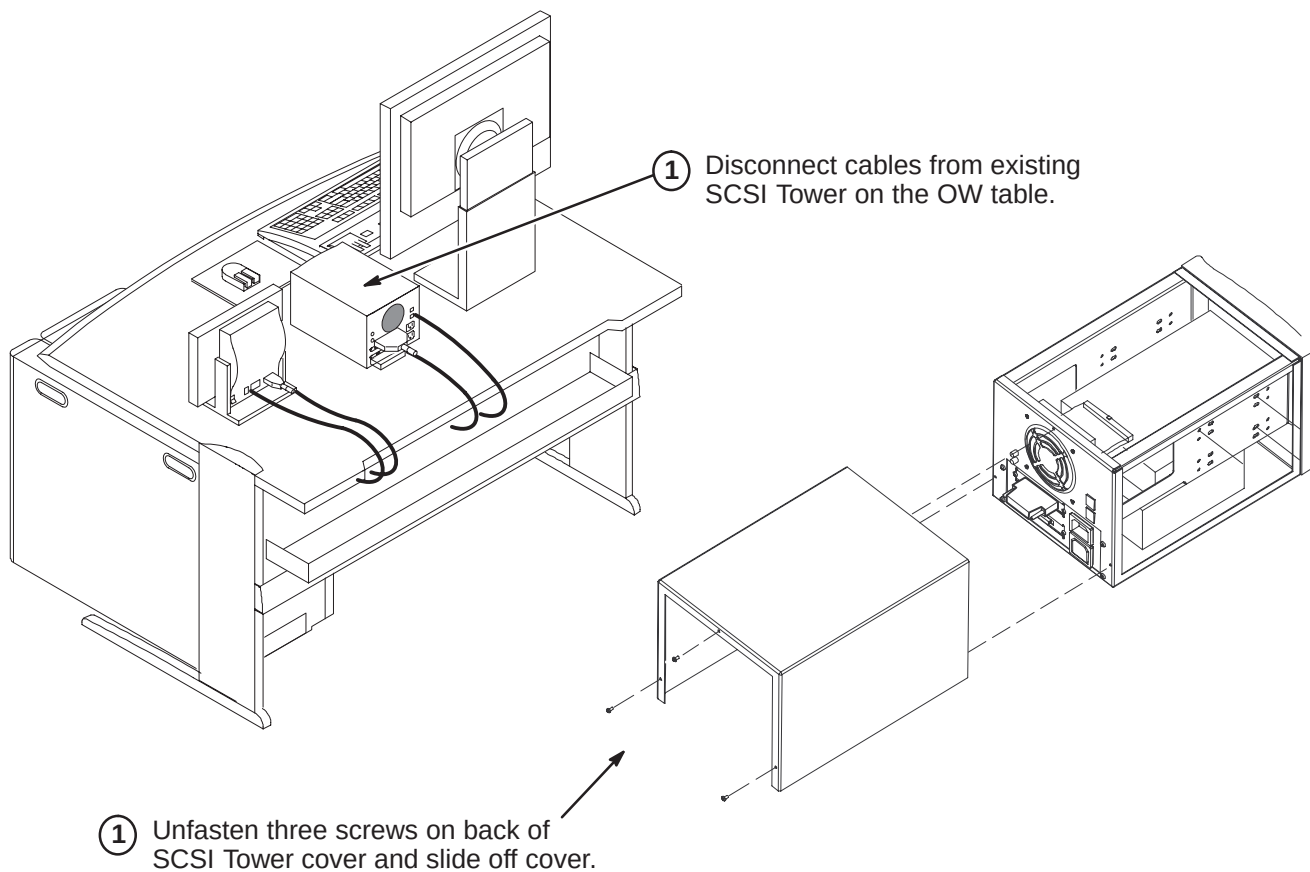
8-1 INTRODUCTION

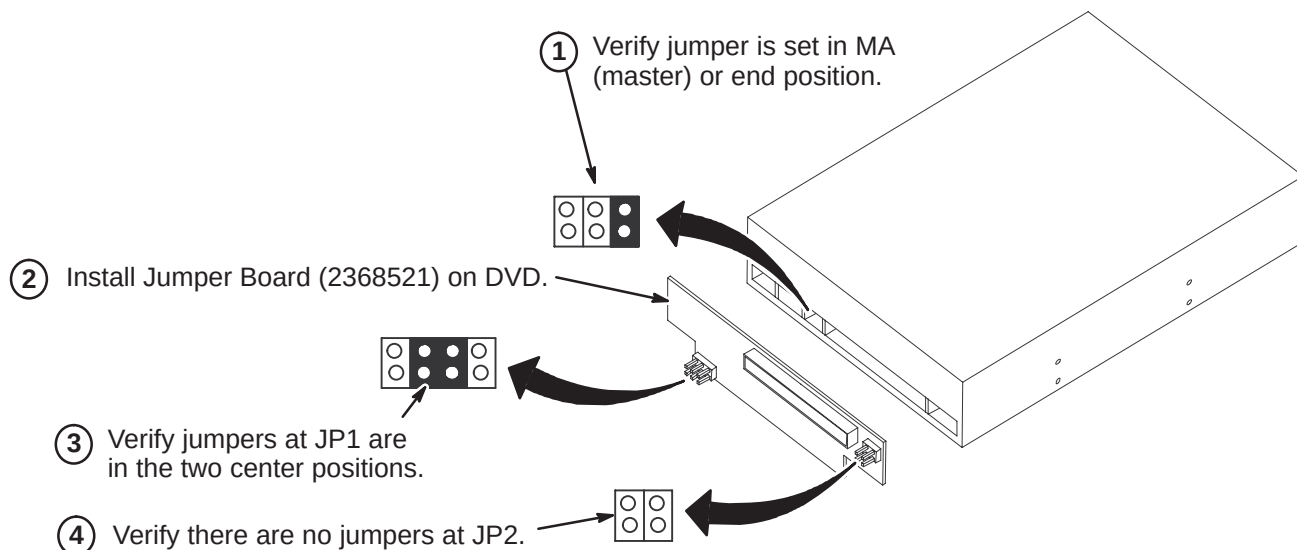
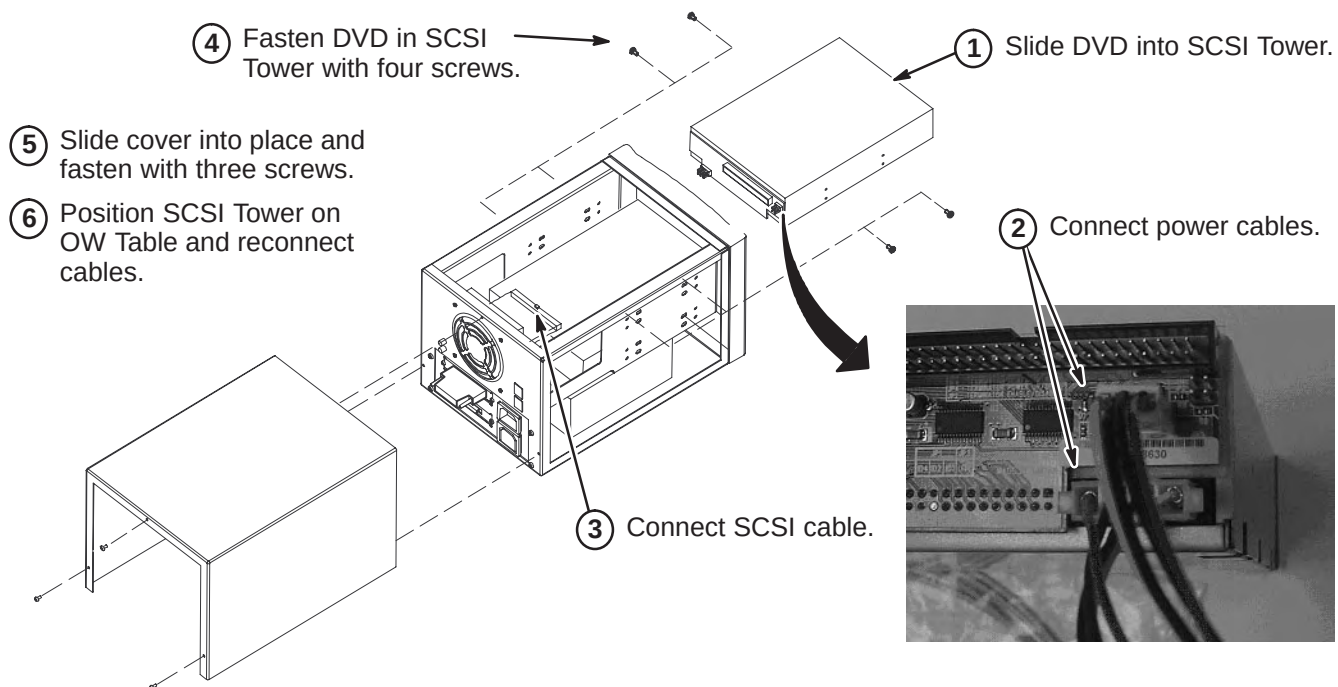
A new DVD-RW will be installed in the existing SCSI Tower on the Operator Workspace Table.

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

8-2 PREPARE SCSI TOWER FOR UPGRADE



8-3 VERIFY JUMPER POSITIONS ON DVD**8-4 INSTALL DVD INTO SCSI TOWER**

SECTION 9 – SYSTEM CABINET

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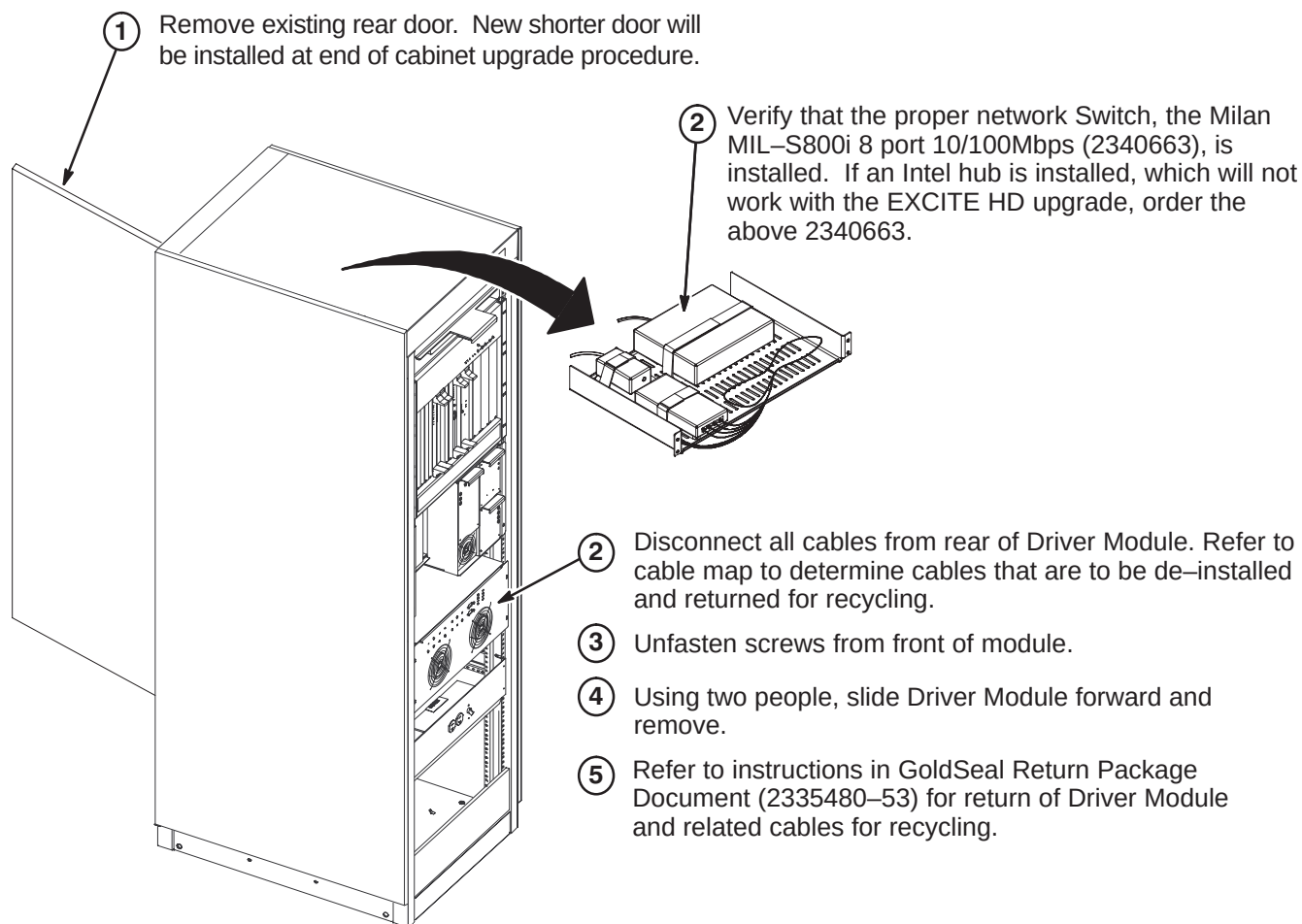
<u>SECTION</u>	<u>PAGE</u>
9-1 INTRODUCTION	9-2
9-2 PREPARE SYSTEM CABINET FOR UPGRADE	9-2
9-3 INSTALL NEW DRIVER MODULE	9-3
9-4 INSTALL COAX POWER SPLITTER	9-3
9-5 INSTALL NEW INTERFACE PANEL	9-4
9-6 PROCEDURES FOR UPGRADING TO 8 CHANNEL	9-5
9-6-1 INSTALL REFLEX 400 AP BOARDS	9-5
9-6-2 REMOVE EXISTING DRF1 AND IRF1 BOARDS IN MGD CHASSIS	9-5
9-6-3 INSTALL CAN TERMINATOR, DRF2, AND IRF2 BOARDS IN MGD CHASSIS	9-6
9-7 PROCEDURES FOR UPGRADING TO 16 CHANNEL	9-7
9-7-1 INSTALL REFLEX 800 AP BOARDS	9-7
9-7-2 REMOVE EXISTING DRF1 AND IRF1 BOARDS IN MGD CHASSIS	9-8
9-7-3 INSTALL CAN TERMINATOR, DRF2, AND IRF2 BOARDS IN MGD CHASSIS	9-8
9-7-4 INSTALL UTNS3, RCVR2, AND EXCR2 BOARDS	9-9
9-7-5 REMOVE BLANK PANELS AND POWER SUPPLY FROM RRF CHASSIS	9-9
9-7-6 INSTALL MODULES IN RRF CHASSIS	9-10
9-7-7 INSTALL CABLES AT REAR OF RRF CHASSIS	9-10
9-7-8 INSTALL CABLES AND POWER SUPPLY AT FRONT OF RRF CHASSIS	9-11
9-8 INSTALL NEW REAR DOOR AND RATING PLATE	9-11

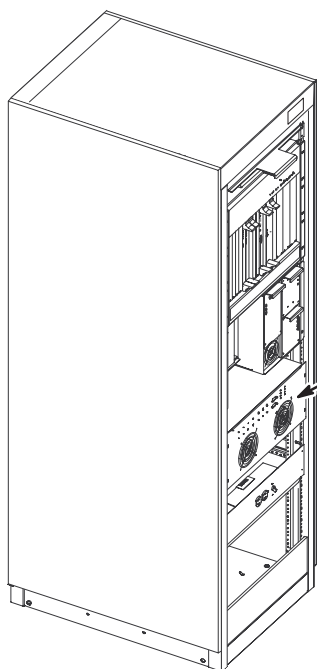
CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

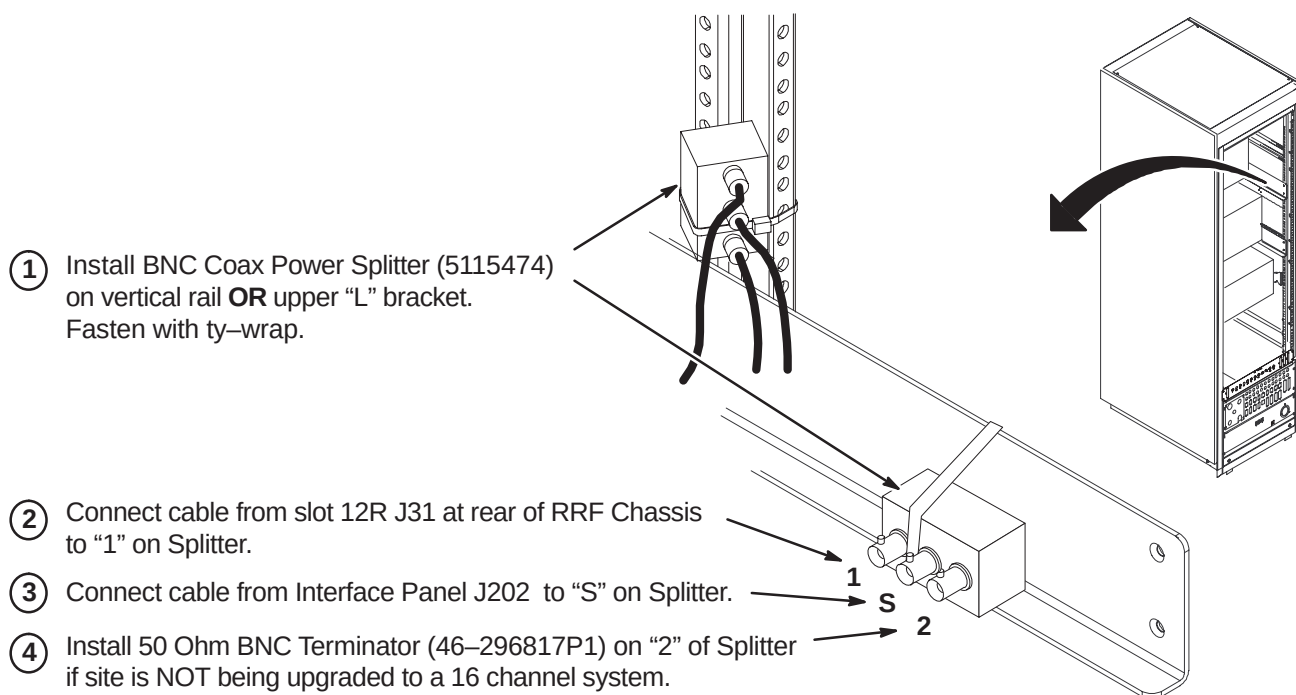
9-1 INTRODUCTION

The System cabinet will be modified for the upgrade to EXCITE Release 12.x. Refer to cable maps in Section 4 for disconnection and removal of existing cables and installation of new cables. Refer to the following steps for the upgrade procedures that pertain to your site.

9-2 PREPARE SYSTEM CABINET FOR UPGRADE

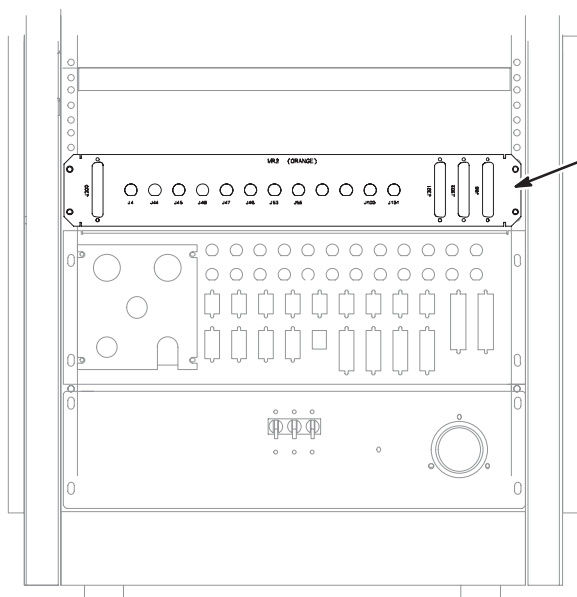
9-3 INSTALL NEW DRIVER MODULE

- ① Using two people, slide new Driver Module into position.
- ② Secure in place with previously removed screws.
- ③ Disconnect all cables from rear of Driver Module. Refer to appropriate cable map to determine cables that are to be removed and returned for recycling.

9-4 INSTALL COAX POWER SPLITTER

- ① Install BNC Coax Power Splitter (5115474) on vertical rail **OR** upper "L" bracket. Fasten with ty-wrap.
- ② Connect cable from slot 12R J31 at rear of RRF Chassis to "1" on Splitter.
- ③ Connect cable from Interface Panel J202 to "S" on Splitter.
- ④ Install 50 Ohm BNC Terminator (46-296817P1) on "2" of Splitter if site is NOT being upgraded to a 16 channel system.

If 16 Channel is installed, connect cable from slot 3R J31 at rear of RRF Chassis to "2" on Splitter.

9-5 INSTALL NEW INTERFACE PANEL

① At rear of cabinet, install 1.5T upgrade Interface Panel (5119238) above existing interface panel and fasten with four (4) M5 x 16mm Pan head screw (2279742-2).

② Use table below for connecting the remaining cables in the System Cabinet. All the cables connected to the new I/O Panel are pre-installed on the I/O Panel.

CABLE PART #	ONE END	OTHER END
5117295	Driver Module J4	MGD Slot 11R J21
5117963	Driver Module J10	I/O Panel J66
2394870-3	RF Slot 14R J5	I/O Panel J200
2394870-4	RF Slot 5R J5	I/O Panel J201
5119896	Splitter S RF Slot 14R J13	I/O Panel J202

9-6 PROCEDURES FOR UPGRADING TO 8 CHANNEL

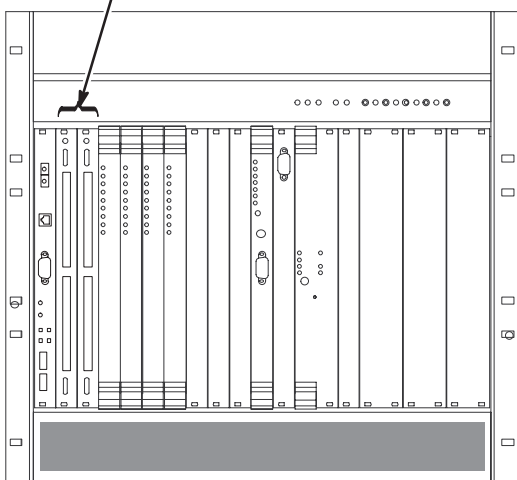
After upgrade, all systems will have one DRF2 board and either one or two IRF2 boards.

9-6-1 INSTALL REFLEX 400 AP BOARDS

Upgrade to an 8 Channel system requires the installation of ReFlex 400 AP boards, M3000RM option. Refer to Direction 5121980, also part of M3000RM, for instructions on installing AP boards.

Note: If site already has ReFlex 400 AP boards installed, then do not perform this procedure.

- ① Remove blank panels or existing non-ReFlex 400B boards.
- ② Install new Master (2325947-12) and Slave (2325847-13). These boards are part of M3000RM.



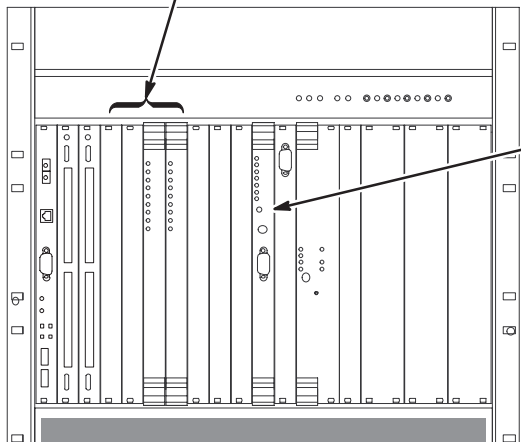
Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

9-6-2 REMOVE EXISTING DRF1 AND IRF1 BOARDS IN MGD CHASSIS



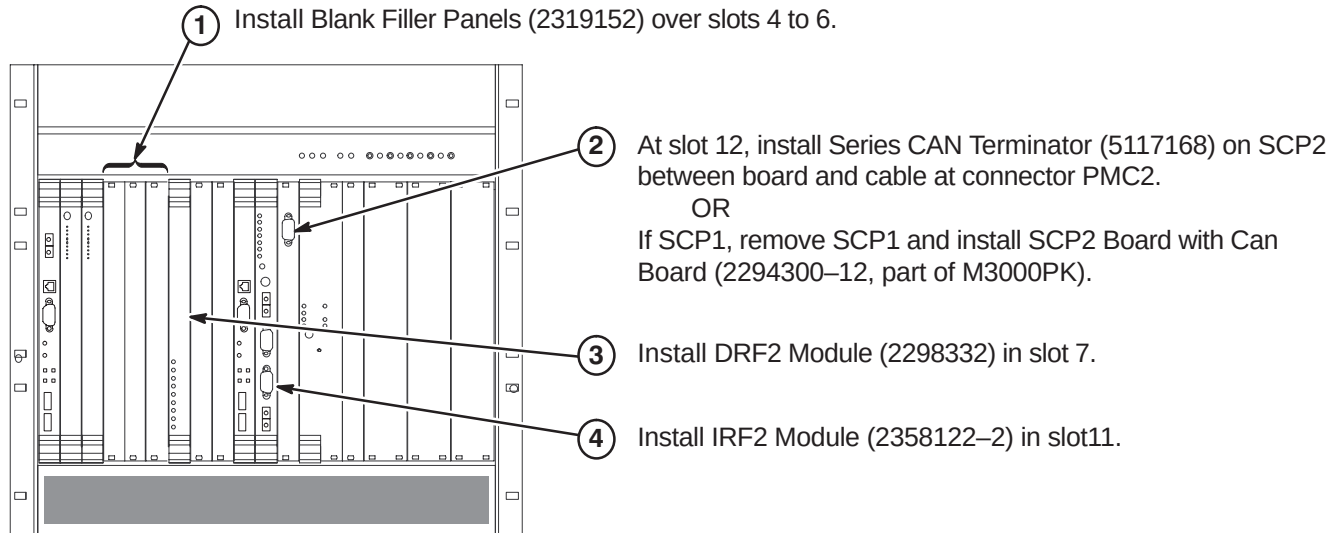
Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

- ① Remove the two (or four) DRF1 boards in slot 4 thru 7. Loosen the screws in the ejector pins, then push the top and bottom tabs to eject the boards from the chassis.
- ② Remove the IRF1 board from slot 11. Loosen the screws in the ejector pins, then push the top and bottom tabs to eject the boards from the chassis.



9-6-3 INSTALL CAN TERMINATOR, DRF2, AND IRF2 BOARDS IN MGD CHASSIS

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).



9-7 PROCEDURES FOR UPGRADING TO 16 CHANNEL

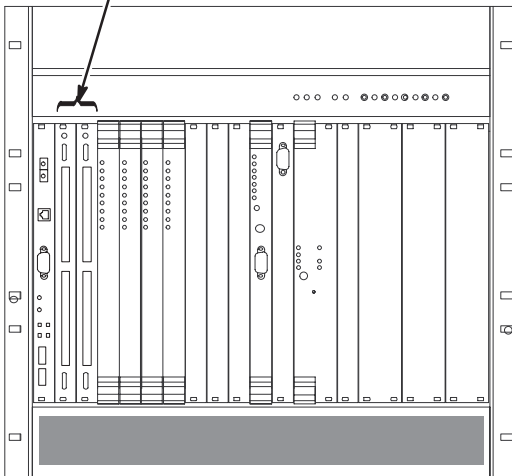
After upgrade, all systems will have one DRF2 board and either one or two IRF2 boards.

Do NOT perform this procedure (9-7) if site is not upgrading to 16 channels.

9-7-1 INSTALL REFLEX 800 AP BOARDS

Upgrade to a 16 Channel system requires the installation of ReFlex 800 AP boards, M3000RN option. Refer to Direction 5121980, also part of M3000RN, for instructions on installing AP boards.

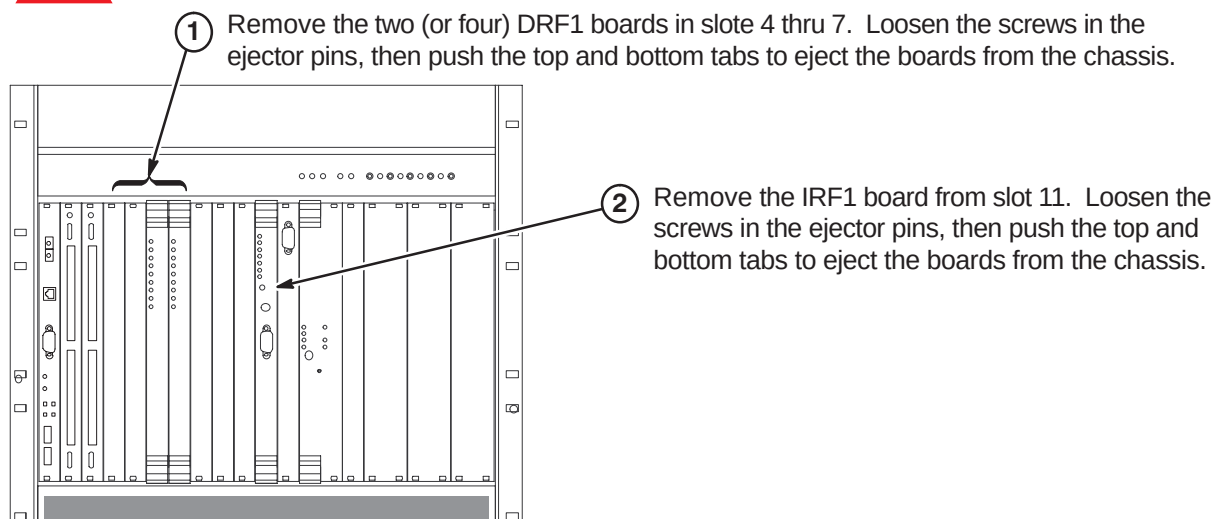
- ① Remove blank panels or existing non-ReFlex 800 boards.
- ② Install new Master (2325947-15) and Slave (2325847-16). These boards are part of M3000RN.



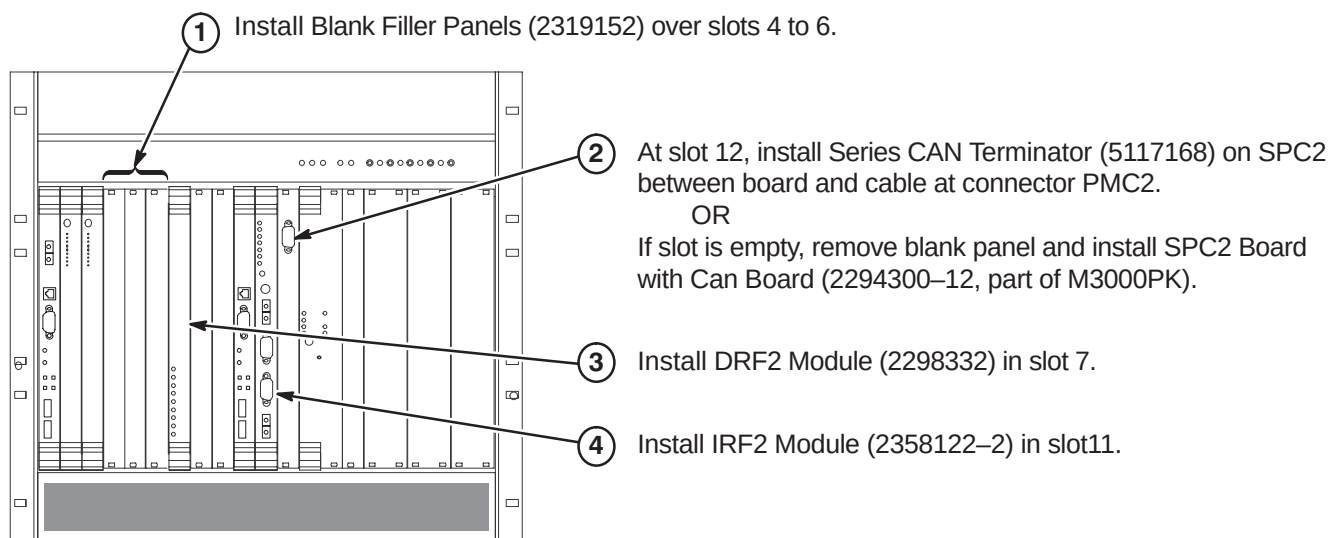
Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

9-7-2 REMOVE EXISTING DRF1 AND IRF1 BOARDS IN MGD CHASSIS

Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

**9-7-3 INSTALL CAN TERMINATOR, DRF2, AND IRF2 BOARDS IN MGD CHASSIS**

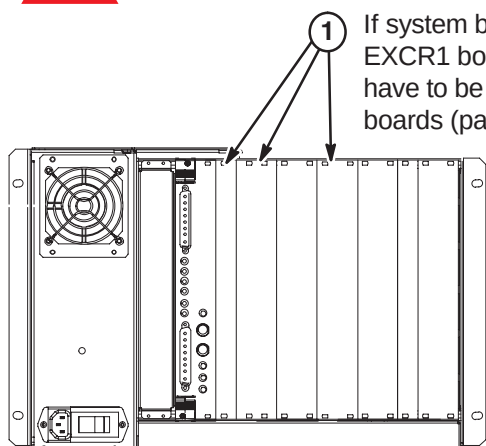
Note: Module is a static sensitive component and needs to be handled as such (wrist strap).



9-7-4 INSTALL UTNS3, RCVR2, AND EXCR2 BOARDS



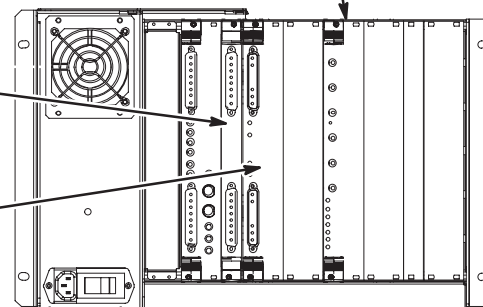
Note: Module is a static sensitive component and needs to be handled as such (wrist strap).



- ① If system being upgraded has UTNS2, RCVR1, and EXCR1 boards at these slots, then these three boards have to be replaced with UTNS3, RCVR2, and EXCR2 boards (part of M3000PH).

- ③ Remove UTNS 2 board and install UTNS 3 board (2366218-2) in slot 12.
- ④ Remove existing RCVR board and install RCVR 2 board (2349808-2) in slots 10 & 11.

- ② Remove existing Exciter board and install Remote RF Master Exciter 2 board (2361470-2) in slots 6 & 7.

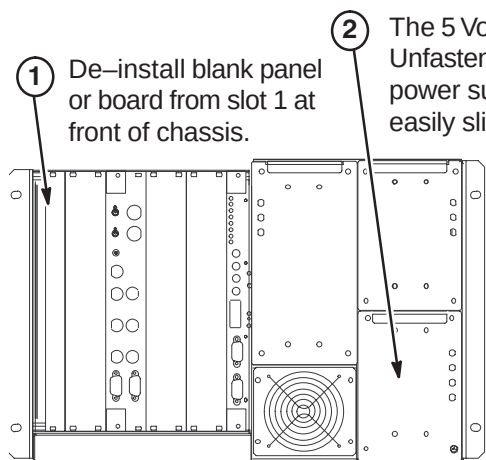


9-7-5 REMOVE BLANK PANELS AND POWER SUPPLY FROM RRF CHASSIS

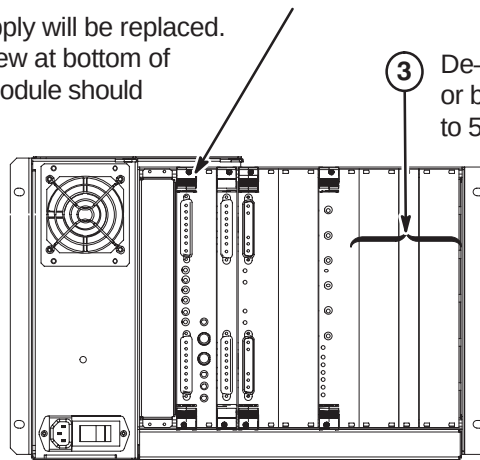


Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

Note: It is recommended that the existing MUX board be at the same part number and revision level as the new MUX board being installed in slots 4 & 5.



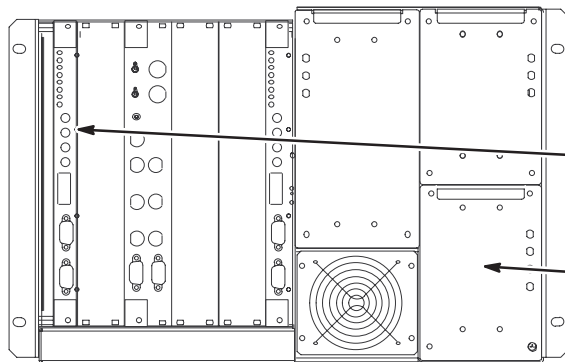
FRONT VIEW



REAR VIEW

Note: All de-installed boards are to be returned for recycling.

9-7-6 INSTALL MODULES IN RRF CHASSIS

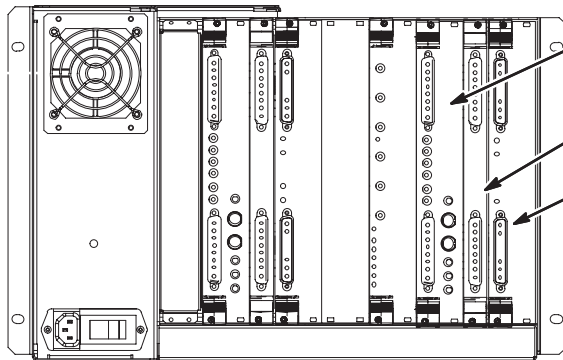


Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

① Install RF-DIF Board (2274184 or 5250028) in slot 1.

② Slide in new 5V RF Module RF Power Supply and secure with four screws previously removed.

FRONT



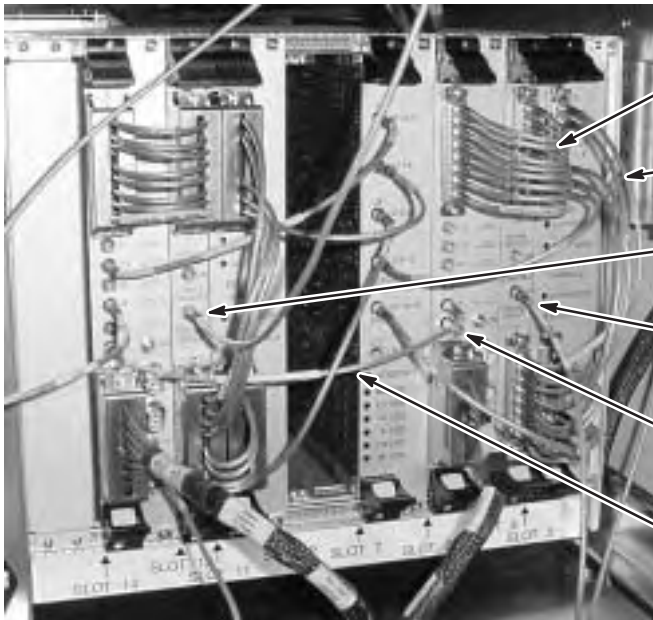
③ Install MUX Board (5250106) in slots 4 and 5.

④ Install UTNS3 Board (2366218-2) in slot 3.

⑤ Install RCVR II Board (2349808-2) in slots 1 and 2.

REAR

9-7-7 INSTALL CABLES AT REAR OF RRF CHASSIS



① Connect cable 5123706 as marked.

② Connect cable 5113193-2 as marked.

③ Connect cable 5115417 slot 12R J31 as marked. Route and connect to Coax Power Splitter "1".

④ Connect cable 5115417-2 as marked. Route and connect to Coax Power Splitter "2".

⑤ Move 50 Ohm Terminator from Mux1 J6 (slots 13 & 14) to Mux2 J6 (slots 4 & 5).

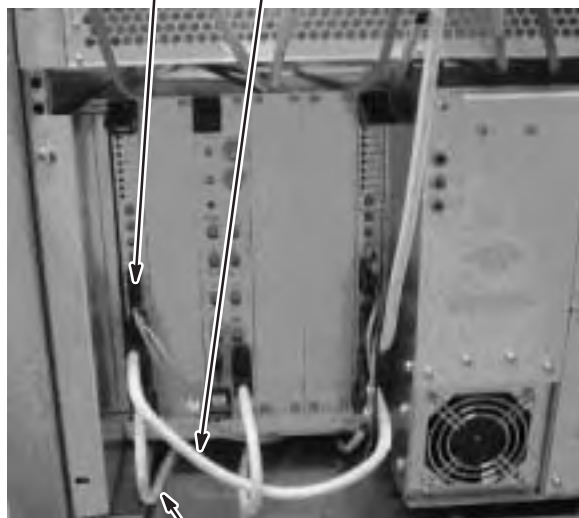
⑥ Connect cable 5115630 as marked.

9-7-8 INSTALL CABLES AND POWER SUPPLY AT FRONT OF RRF CHASSIS

③ Connect cable 5121122-2 as marked to RF Slot 1F J21. Route and connect other end to MGD Slot 11F J7.

② Connect cable 5113182-2 as marked.

④ Slide in new 5V Power Supply (2292267-2). Make sure that it is properly seated before tightening the four screws.



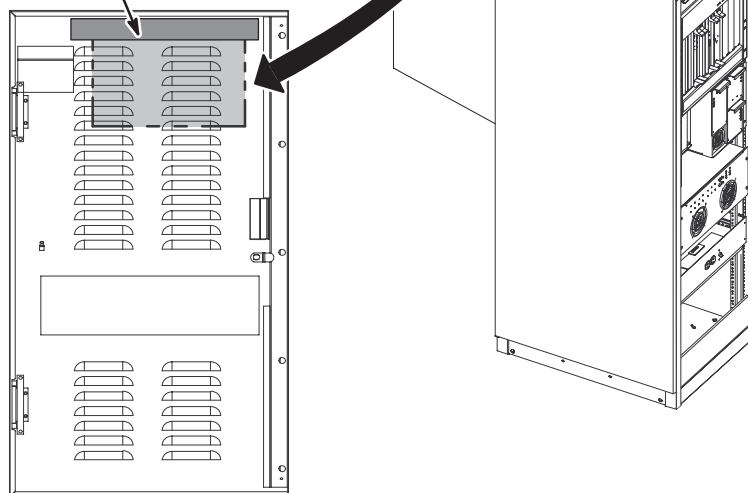
① Disconnect cable (2323878-3) from Slot 10F J23 and connect to Slot 1F J23.

**9-8 INSTALL NEW REAR DOOR AND RATING PLATE**

① Install new rear System Cabinet door (5115834).

③ Attach upgrade Rating Plate (46-302200P12) to top of cabinet on back side.

② At top of inside of rear door, attach plastic bag containing Service Cable Kit (5117087) with filament tape.



SECTION 10 – COMPLETION

10-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 10-2, 10-3, and 10-4.

10-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 10-3. Recycling Operation will route them to appropriate location.

10-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-7. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

10-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1-8.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 12.x has been made before returning system to the users. Turn site over to applications who will instruct users.

10-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
 - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the ACGD Gradient power modules, switch all cabinet circuit breakers to ON. The ACGD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.